

Project FORESIGHT

AI-Powered Demand & Inventory Intelligence Platform

Notebook 02: Exploratory Data Analysis

Project Information

Item	Details
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1. Business Context and Notebook Objective

Retail demand changes across time, products, stores, prices, events, and seasonal periods. Without understanding these patterns, forecasting models and inventory recommendations may produce unreliable or misleading results.

Notebook 02 analyses the validated datasets created in Notebook 01 to identify the main demand, sales, pricing, product, store, calendar, and simulated inventory patterns within the 300 SKU-store development scope.

Objectives

This notebook will:

- validate and integrate the processed project datasets;
- summarise the project's main sales and inventory KPIs;
- analyse demand trends and seasonality;
- compare category, department, store, state, product, and SKU performance;
- identify top movers, slow movers, and dead-stock candidates;
- examine price, event, SNAP, and simulated inventory relationships;

- translate analytical findings into clear business insights and recommendations;
- export validated tables and figures for reporting and downstream project stages.

Important: Inventory values used in this notebook are simulated and must be interpreted as analytical estimates rather than observed retailer stock records.

2. Notebook Workflow and Business Questions

Notebook 02 follows a focused exploratory workflow designed to understand historical demand, identify business risks and opportunities, and prepare reliable analytical outputs for feature engineering and forecasting.

2.1 Notebook Workflow

Section	Focus
3	Notebook Setup
4	Load Processed Data and Confirm Input Integrity
5	Build the Integrated EDA Dataset
6	Executive KPI Overview
7	Demand Trends and Seasonality
8	Business Performance Analysis
9	Top Movers, Slow Movers, and Dead-Stock Candidates
10	Demand Drivers
11	Inventory and Demand Behaviour
12	Business Insights, Recommendations, and EDA Insight Memo
13	Export and Final Validation
14	Notebook Conclusion

2.2 Core Business Questions

This notebook will answer:

1. How has demand changed over time?
2. When does demand usually increase or decrease?
3. Which categories, departments, stores, states, products, and SKUs perform best?
4. Which products are fast-moving, slow-moving, or possible dead-stock candidates?
5. How are price, events, promotions, and SNAP periods associated with demand?

6. Which products show stable, volatile, or intermittent demand?
7. Where does simulated inventory appear insufficient or excessive?
8. Which findings should influence forecasting and inventory planning?

2.3 Analytical Principles

- Use only the validated 300 SKU-store development scope.
- Avoid repeating the detailed engineering checks completed in Notebook 01.
- Create charts only when they answer a defined business question.
- Separate observed M5 data from simulated inventory information.
- Interpret relationships as associations unless causal evidence exists.
- Convert analytical findings into concise business insights and recommendations.

Scope boundary: This notebook performs exploratory analysis only. Forecast features, model training, final risk scoring, and reorder calculations belong to later notebooks.

3. Notebook Setup

This section prepares a consistent and reproducible environment for exploratory analysis. It defines notebook settings, imports the required libraries, and establishes reusable project paths.

Objectives

- configure the notebook environment;
- import libraries for data manipulation and visualisation;
- define processed-data and output directories;
- prepare consistent display and chart settings;
- confirm that the environment is ready for analysis.

3.1 Notebook Configuration

Non-critical warnings and notebook-wide display settings will be configured to keep outputs readable while preserving important validation messages.

✅ Section 3.1 completed: Notebook configuration applied.

3.2 Import Required Libraries

The required libraries will support:

- file and directory management;
- data manipulation;

- numerical analysis;
- statistical summaries;
- data visualisation;
- enhanced notebook display.

✅ Section 3.2 completed: Required libraries imported successfully.

Library Summary

The required Python libraries were successfully imported and configured for Notebook 02. Each library supports a specific part of the exploratory data analysis workflow.

Library	Purpose
pandas	Data loading, manipulation, aggregation, and analysis
NumPy	Numerical calculations and statistical operations
Matplotlib	Creation of analytical charts and figures
Seaborn	Statistical visualisation and consistent chart styling
Pathlib	Project directory and file-path management
IPython.display	Enhanced display of tables and notebook outputs

The notebook environment is now prepared for data loading, exploratory analysis, visualisation, and output generation.

3.3 Project Directories and File Paths

Reusable paths will be defined for:

- processed input datasets;
- interim analytical datasets;
- exported summary tables;
- analytical figures;
- executive reports;
- notebook exports.

Using centralised paths avoids machine-specific file references and supports reproducibility.

✅ Section 3.3 completed: Project paths configured successfully.

```
Project root      : C:\Users\hp\Project FORESIGHT
Processed data    : C:\Users\hp\Project FORESIGHT\data\processed
Interim data      : C:\Users\hp\Project FORESIGHT\data\interim
Reports directory : C:\Users\hp\Project FORESIGHT\reports
```

Directory Configuration Summary

The Project FORESIGHT root directory was identified correctly as:

C:\Users\hp\Project FORESIGHT

The processed-data, interim-data, and reporting paths point to the expected project folders. Centralising these paths ensures that Notebook 02 can load inputs and export outputs without using repeated or machine-specific file references.

✔ Project directories and file paths are configured correctly.

3.4 Output Directory Preparation

Required output directories will be created through code where necessary.

This ensures that later exports do not fail because a destination folder is missing.

	Directory	Path	Exists
0	Interim Data	C:\Users\hp\Project FORESIGHT\data\interim	True
1	Summary Tables	C:\Users\hp\Project FORESIGHT\reports\tables	True
2	Analytical Figures	C:\Users\hp\Project FORESIGHT\reports\figures	True
3	Executive Reports	C:\Users\hp\Project FORESIGHT\reports\executiv...	True
4	Notebook Exports	C:\Users\hp\Project FORESIGHT\reports\notebook...	True

✔ Section 3.4 completed: Output directories are ready.

✔ Notebook Setup Summary

Notebook 02 is now configured for reproducible exploratory analysis.

The following setup tasks were completed:

- non-critical warnings were suppressed;
- required analytical and visualisation libraries were imported;
- centralised project paths were defined;
- processed-data and output locations were configured;
- all required output directories were confirmed successfully.

The notebook environment is ready to load the validated datasets produced by Notebook 01.

✔ Section 3 is complete.

4. Load Processed Data and Confirm Input Integrity

This section loads the validated datasets produced by Notebook 01 and performs a small set of handoff checks before exploratory analysis begins.

The detailed data-quality and engineering validation has already been completed in Notebook 01. Therefore, only checks required to protect the Notebook 02 workflow will be performed here.

Input Datasets

- sales_daily_final.csv
- calendar_final.csv
- sku_master_final.csv
- inventory_snapshots_final.csv
- development_sku_scope.csv

Objectives

- confirm that all required files are available;
- load the processed datasets;
- verify expected key columns;
- confirm the 300 SKU-store development scope;
- review dataset dimensions and date coverage;
- confirm readiness for analytical integration.

4.1 Confirm Required Input Files

The notebook will verify that each required processed file exists and is not empty before loading it.

	Dataset	Filename	Exists	Size (MB)	Path
0	Sales Daily	sales_daily_final.csv	True	24.41	C:\Users\hp\Project FORESIGHT\data\processed\s...
1	Calendar	calendar_final.csv	True	0.15	C:\Users\hp\Project FORESIGHT\data\processed\c...
2	SKU Master	sku_master_final.csv	True	3.66	C:\Users\hp\Project FORESIGHT\data\processed\s...
3	Inventory Snapshots	inventory_snapshots_final.csv	True	42.30	C:\Users\hp\Project FORESIGHT\data\processed\i...
4	Development Scope	development_sku_scope.csv	True	0.01	C:\Users\hp\Project FORESIGHT\data\processed\d...

✔ Section 4.1 completed: All required input files are available.

Input File Summary

All five required Notebook 02 input files were found successfully in the processed-data directory.

Dataset	File Size
Sales Daily	24.41 MB
Calendar	0.15 MB
SKU Master	3.66 MB
Inventory Snapshots	42.30 MB
Development Scope	0.01 MB

The files are available and non-empty, so the notebook can proceed to load the validated datasets.

✓ Required input-file handoff check passed.

4.2 Load Processed Datasets

The validated datasets will be loaded into Pandas DataFrames without repeating the cleaning and engineering steps completed in Notebook 01.

	Dataset	Rows	Columns	Memory (MB)
0	Sales Daily	573900	6	36.12
1	Calendar	1969	22	0.38
2	SKU Master	30490	16	6.29
3	Inventory Snapshots	573900	17	88.69
4	Development Scope	300	1	0.01

✓ Section 4.2 completed: All processed datasets loaded successfully.



Dataset Loading Summary

The four core business datasets and one internal scope-control file were loaded successfully.

File Type	Dataset	Rows	Columns
Core Dataset	Sales Daily	573,900	6
Core Dataset	Calendar	1,969	22
Core Dataset	SKU Master	30,490	16
Core Dataset	Inventory Snapshots	573,900	17
Control File	Development Scope	300	1

The four core datasets match the structure required for Project FORESIGHT. The development-scope file is used only to keep all notebooks restricted to the same approved 300 SKU-store series.

The `date` columns in the sales, calendar, and inventory datasets were loaded as datetime values, and no file loaded as empty.

✓ Core datasets and scope control are ready for the analytical-input integrity check.

4.3 Confirm Analytical Inputs

A concise integrity check will confirm:

- expected dataset dimensions;
- required date and identifier columns;
- 300 unique development-scope SKUs;

- available historical date range;
- readiness for integration.

Note: This is an input handoff check, not a repetition of the complete Notebook 01 validation process.

Analytical-input integrity framework prepared.
Datasets registered : 5

Structural integrity checks prepared.
Checks recorded : 9

	Check	Status	Details
0	Sales Daily: required columns	Passed	All required columns found
1	Calendar: required columns	Passed	All required columns found
2	SKU Master: required columns	Passed	All required columns found
3	Inventory Snapshots: required columns	Passed	All required columns found
4	Development Scope: required columns	Passed	All required columns found
5	Development scope contains 300 unique SKU IDs	Passed	Rows: 300; Unique SKU IDs: 300; Duplicates: 0
6	Sales Daily: development-scope coverage	Passed	All 300 scope SKU IDs found
7	SKU Master: development-scope coverage	Passed	All 300 scope SKU IDs found
8	Inventory Snapshots: development-scope coverage	Passed	All 300 scope SKU IDs found
9	Sales date column is datetime	Passed	datetime64[us]
10	Calendar date column is datetime	Passed	datetime64[us]
11	Inventory date column is datetime	Passed	datetime64[us]
12	Sales analytical key is unique	Passed	Duplicate sku_id-date keys: 0
13	Inventory analytical key is unique	Passed	Duplicate sku_id-date keys: 0
14	Sales historical coverage is available	Passed	2011-01-29 to 2016-04-24 (1,913 unique dates)

ANALYTICAL INPUT INTEGRITY SUMMARY

=====

Total checks : 15
Passed : 15
Failed : 0

- ✓ Section 4.3 completed: Analytical inputs confirmed.
- ✓ Four core datasets and the scope-control file are ready.

✓ Analytical Input Integrity Summary

All analytical-input integrity checks passed successfully.

The validation confirmed that:

- all required key columns are available;
- the development-scope file contains 300 unique and non-null SKU-store identifiers;
- all 300 scope identifiers are present in the Sales Daily, SKU Master, and Inventory Snapshot datasets;
- the sales, calendar, and inventory date fields use valid datetime types;
- the `sku_id + date` key is unique in the sales and inventory datasets;
- historical sales coverage is available for exploratory analysis.

The four core business datasets and one internal scope-control file are ready for integration.

✓ Section 4 is complete.

5. Build the Integrated EDA Dataset

This section combines the processed project datasets into one analysis-ready table for exploratory analysis.

The integrated dataset will retain a clear daily SKU-store grain while adding calendar, product, pricing, and simulated inventory information.

Objectives

- define the analytical grain;
 - restrict the analysis to the approved 300 SKU-store scope;
 - integrate sales, calendar, SKU, and inventory attributes;
 - prevent duplicate or multiplied records during merging;
 - confirm row-count and key consistency;
 - prepare a reusable analytical dataset for the remaining EDA sections.
-

5.1 Define the Analytical Grain

The integrated EDA dataset will contain one row per SKU-store-date.

The expected composite key is:

```
sku_id + date
```

Analytical grain defined successfully.

Composite key : sku_id + date

SKU-store series : 300

Historical dates : 1,913

	Dataset	Grain	Rows	Unique SKU IDs	Unique Dates	Duplicate Keys
0	Sales Daily	One row per SKU-store-date	573900	300	1913	0
1	Inventory Snapshots	One row per SKU-store-date	573900	300	1913	0
2	SKU Master	One row per SKU-store	30490	30490	Not applicable	0
3	Calendar	One row per date	1969	Not applicable	1969	0

✅ Analytical-grain validation passed.

ANALYTICAL GRAIN DEFINITION

```
=====
Daily analytical grain : One row per SKU-store-date
Composite key          : sku_id + date
Development scope      : 300 SKU-store series
Historical dates       : 1,913
Expected daily rows    : 573,900
```

✅ Section 5.1 completed: Analytical grain confirmed.



Analytical Grain Summary

The core datasets follow the expected analytical structure:

Dataset	Grain	Key Validation
Sales Daily	One row per SKU-store-date	573,900 unique daily records
Inventory Snapshots	One row per SKU-store-date	573,900 unique daily records
SKU Master	One row per SKU-store	30,490 unique SKU-store records
Calendar	One row per date	1,969 unique dates

For Notebook 02, the integrated analytical dataset will use:

- **Grain:** one row per SKU-store-date;
- **Composite key:** `sku_id + date` ;
- **Development scope:** 300 SKU-store series;
- **Historical coverage:** 1,913 sales dates;
- **Expected analytical rows:** 573,900.

No duplicate keys were detected in any core dataset.

✅ The analytical grain is clearly defined and ready for scope application.

5.2 Apply the Development Scope

The sales, SKU, and inventory records will be restricted to the 300 approved SKU-store identifiers listed in `development_sku_scope.csv` .

Using one consistent scope ensures that all KPIs, charts, and exported summaries describe the same analytical population.

Development scope prepared successfully.
Approved SKU-store series : 300

	Dataset	Original Rows	Scoped Rows	Scope SKU IDs	Rows Removed
0	Sales Daily	573900	573900	300	0
1	SKU Master	30490	300	300	30190
2	Inventory Snapshots	573900	573900	300	0

✅ Development-scope validation passed.

DEVELOPMENT SCOPE SUMMARY

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Approved SKU-store series : 300
 Historical dates : 1,913
 Expected daily rows : 573,900
 Scoped SKU Master rows : 300

✅ Section 5.2 completed: Development scope applied.

Development Scope Summary

The approved 300 SKU-store development scope was applied consistently across all SKU-level datasets.

Dataset	Scoped Rows	Unique SKU IDs	Result
Sales Daily	573,900	300	Already within scope
SKU Master	300	300	Filtered from the complete master
Inventory Snapshots	573,900	300	Already within scope

The Sales Daily and Inventory Snapshot datasets were created using the approved development scope in Notebook 01, so no rows were removed from them. The complete SKU Master contained all 30,490 available SKU-store combinations and was filtered to the corresponding 300 records required for Notebook 02.

The complete `sku_master_final.csv` remains preserved as the canonical master dataset, while `sku_master_scope` is the 300-row working version used for integration.

✅ The development scope is consistent and ready for dataset integration.

5.3 Integrate the Processed Datasets

The datasets will be combined using their validated business keys:

- sales and calendar through `date` ;
- sales and SKU master through `sku_id` ;
- sales and inventory snapshots through `sku_id` and `date` .

Only fields required for EDA, reporting, and later feature engineering will be retained.

Scoped sales base prepared successfully.

Rows : 573,900

Columns : 6

Calendar attributes integrated successfully.

Columns added : 21

Unmatched rows : 0

SKU Master attributes integrated successfully.

Columns added : 15

Unmatched rows : 0

Inventory attributes integrated successfully.

Columns added : 14

Unmatched rows : 0

	Integration Stage	Rows	Columns	Columns Added	Unmatched Rows
0	Scoped Sales Base	573900	6	6	0
1	Calendar Added	573900	27	21	0
2	SKU Master Added	573900	42	15	0
3	Inventory Added	573900	56	14	0

✅ Dataset-integration validation passed.

DATASET INTEGRATION SUMMARY

=====

Integrated rows : 573,900

Integrated columns : 56

Unique SKU-store IDs : 300

Historical dates : 1,913

Duplicate keys : 0

✅ Section 5.3 completed: Processed datasets integrated.



Dataset Integration Summary

The four processed business datasets were integrated successfully at the defined daily analytical grain.

Integration Stage	Rows	Columns Added	Unmatched Rows
Scoped Sales Base	573,900	6	0
Calendar Added	573,900	21	0
SKU Master Added	573,900	15	0
Inventory Added	573,900	14	0

The integration preserved all 573,900 sales records and produced a unified dataset containing 56 columns.

The final integrated dataset contains:

- 300 SKU-store series;
- 1,913 historical dates;
- no unmatched calendar, SKU Master, or inventory records;

- no duplicate `sku_id + date` keys;
- no unexpected row multiplication during merging.

✓ The processed datasets are integrated and ready for merge-integrity validation.

5.4 Confirm Merge Integrity

After integration, concise checks will confirm:

- expected row count;
- uniqueness of the analytical key;
- absence of unexpected row multiplication;
- successful matching of essential attributes;
- preservation of the historical date range.

Merge-integrity framework prepared.

Expected rows : 573,900

Expected columns : 56

Core merge-integrity checks prepared.

Checks recorded : 6

Panel and column checks prepared.

Checks recorded : 6

	Check	Status	Details
0	Integrated row count preserved	Passed	Expected: 573,900; Found: 573,900
1	Composite analytical key is unique	Passed	Duplicate sku_id-date keys: 0
2	Composite analytical key is complete	Passed	Missing key values: 0
3	Approved development scope preserved	Passed	Expected SKU IDs: 300; Found: 300
4	Historical date coverage preserved	Passed	Expected dates: 1,913; Found: 1,913
5	Historical date range preserved	Passed	2011-01-29 to 2016-04-24
6	Every SKU contains complete daily history	Passed	Expected records per SKU: 1,913; Observed rang...
7	Every date contains all scope SKUs	Passed	Expected SKUs per date: 300; Observed range: 3...
8	All expected source columns are available	Passed	No expected columns missing
9	No unexpected columns were introduced	Passed	No unexpected columns found
10	Integrated column names are unique	Passed	Duplicate column names: 0
11	No automatic merge-suffix columns remain	Passed	No _x or _y columns found
12	units_sold total preserved	Passed	Source: 573,932.00; Integrated: 573,932.00
13	revenue total preserved	Passed	Source: 1,720,316.59; Integrated: 1,720,316.59

MERGE INTEGRITY SUMMARY

```
=====
Total checks : 14
Passed       : 14
Failed       : 0
```

- ✓ Section 5.4 completed: Merge integrity confirmed.
- ✓ The integrated analytical dataset is structurally reliable.

✓ Merge Integrity Summary

The integrated analytical dataset passed all 14 merge-integrity checks.

The validation confirmed that:

- all 573,900 expected records were preserved;
- the `sku_id + date` composite key is complete and unique;
- all 300 approved SKU-store series are retained;
- the full 1,913-day history from 29 January 2011 to 24 April 2016 is preserved;
- every SKU has a complete daily history;
- every date contains all 300 scope identifiers;
- all expected source columns are available;
- no duplicate, unexpected, or automatic merge-suffix columns were introduced;
- total units sold and revenue remain unchanged after integration.

✔ The integrated analytical dataset is complete, consistent, and structurally reliable.

5.5 Prepare and Export the Analytical Base

The validated integrated dataset will be stored as:

`data/interim/analytics_base_daily.parquet`

This dataset will support the remaining Notebook 02 analyses and provide a reusable foundation for downstream project stages.

Important: Inventory-related fields in the integrated dataset are simulated and will remain clearly identified as analytical estimates.

Analytical base prepared successfully.

Rows : 573,900
Columns : 56
SKU-store series : 300
Historical dates : 1,913

✔ Pre-export analytical-base validation passed.

✔ Analytical-base export verification passed.

Exported rows : 573,900
File size : 4.60 MB

	Dataset	Grain	Rows	Columns	Unique SKU IDs	Unique Dates	Duplicate Keys	Memory (MB)	File Size (MB)
0	Analytics Base Daily	One row per SKU-store-date	573900	56	300	1913	0	314.07	4.60

ANALYTICAL BASE EXPORT SUMMARY

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Export path : C:\Users\hp\Project FORESIGHT\data\interim\analytics_base_daily.parquet
Rows : 573,900
Columns : 56
SKU-store series : 300
Historical dates : 1,913
File size : 4.60 MB

- ✔ Section 5.5 completed: Analytical base prepared and exported.
- ✔ Exported file successfully passed the verification checks.

Analytical Base Export Summary

The integrated analytical base was prepared, validated, and exported successfully.

Attribute		Result
Analytical grain	One row per SKU-store-date	
Rows	573,900	
Columns	56	
SKU-store series	300	
Historical dates	1,913	
Duplicate keys	0	
In-memory size	314.07 MB	
Exported file size	4.60 MB	

The dataset was exported to:

`data/interim/analytics_base_daily.parquet`

The final verification confirmed that the exported file exists, is not empty, preserves all expected rows, and retains a unique `sku_id + date` analytical key.

Parquet is a compressed binary format, so it must be viewed using `pd.read_parquet()` rather than Jupyter’s text editor.

- ✔ The analytical base is verified and ready for exploratory data analysis.
- ✔ Section 5 is complete.

6. Executive KPI Overview

This section provides a concise summary of the historical sales, revenue, product coverage, and simulated inventory position within the approved 300 SKU-store development scope.

The KPI overview establishes a common business baseline before detailed trend, product, store, and inventory analyses are performed.

Objectives

- summarise the historical analysis period;
 - calculate headline demand and revenue metrics;
 - measure product, store, category, and geographic coverage;
 - review average demand and pricing indicators;
 - summarise key simulated inventory measures;
 - create an executive-level KPI table for reporting and dashboard development.
-

6.1 Historical Coverage

This subsection will summarise:

- analysis start date;
- analysis end date;
- number of historical days;
- number of available weeks, months, and years;
- number of active SKU-store series.

Historical coverage measures calculated.

Analysis period : 2011-01-29 to 2016-04-24

Historical days : 1,913

SKU-store series: 300

	KPI	Value	Business Meaning
0	Analysis Start Date	29 January 2011	Beginning of the historical analysis period
1	Analysis End Date	24 April 2016	Latest date included in the analysis
2	Historical Days	1,913	Unique daily observations available
3	Observed Calendar Weeks	274	Weekly periods represented in the data
4	Observed Calendar Months	64	Monthly periods represented in the data
5	Observed Calendar Years	6	Calendar years represented in the data
6	Active SKU-Store Series	300	Approved product-store series under analysis

✅ Historical-coverage validation passed.

HISTORICAL COVERAGE SUMMARY

```
=====
Analysis period      : 2011-01-29 to 2016-04-24
Historical days      : 1,913
Observed calendar weeks : 274
Observed calendar months: 64
Observed calendar years : 6
Active SKU-store series : 300
Complete panel rows   : 573,900
```

✅ Section 6.1 completed: Historical coverage confirmed.

Historical Coverage Summary

The approved development scope contains a complete and continuous historical panel from **29 January 2011 to 24 April 2016**.

Coverage Measure	Result
Historical Days	1,913
Observed Calendar Weeks	274
Observed Calendar Months	64
Observed Calendar Years	6
Active SKU-Store Series	300
Complete Panel Rows	573,900

Every SKU-store series contains the full daily history, with no missing dates in the analysis period. The dataset therefore provides sufficient multi-year coverage for trend, seasonality, product-performance, and demand-behaviour analysis.

✓ Historical coverage is complete and suitable for exploratory analysis.

6.2 Sales and Revenue KPIs

This subsection measures the commercial scale and average demand level of the approved 300 SKU-store development scope.

The analysis calculates:

- total units sold and historical revenue;
- average daily and weekly demand;
- weighted average realised selling price;
- estimated gross margin based on the assumption-driven cost and margin fields created in Notebook 01.

Note: Revenue and margin are reported in generic currency units because the source data does not define an official currency. Estimated gross margin is assumption-based and should not be interpreted as observed accounting profit.

Commercial summaries prepared successfully.

Daily periods : 1,913

Weekly periods : 274

✓ Sales and revenue KPI validation passed.

	KPI	Value	Unit	Business Meaning
0	Total Units Sold	573,932	Units	Total historical customer demand
1	Total Revenue	1,720,316.59	Currency Units	Total historical sales value
2	Average Daily Demand	300.02	Units per Day	Average demand across each observed day
3	Average Weekly Demand	2,094.64	Units per Week	Average demand across each observed retail week
4	Weighted Average Selling Price	3.00	Currency Units per Unit	Average realised selling price weighted by uni...
5	Estimated Gross Margin Value	542,429.18	Currency Units	Assumption-based margin using units sold and e...

SALES AND REVENUE KPI SUMMARY

=====

Total units sold : 573,932
 Total revenue : 1,720,316.59
 Average daily demand : 300.02
 Average weekly demand : 2,094.64
 Weighted average sell price : 3.00
 Estimated gross margin value : 542,429.18

✅ Section 6.2 completed: Sales and revenue KPIs calculated.

Sales and Revenue KPI Summary

The approved 300 SKU-store scope generated **573,932 units sold** and **1,720,316.59 currency units in revenue** between January 2011 and April 2016.

Commercial KPI	Result
Total Units Sold	573,932
Total Revenue	1,720,316.59
Average Daily Demand	300.02 units
Average Weekly Demand	2,094.64 units
Weighted Average Selling Price	3.00 per unit
Estimated Gross Margin	542,429.18

Historical demand averaged approximately **300 units per day** and **2,095 units per retail week**. The weighted average selling price of **3.00 per unit** represents the realised average price after accounting for sales volume.

The estimated gross margin is assumption-based and was calculated using the estimated per-unit margin created in Notebook 01. It provides an analytical approximation of commercial value rather than audited accounting profit.

✅ Section 6.2 is complete.

6.3 Product and Business Coverage

This subsection measures the organisational and product coverage represented within the approved 300 SKU-store development scope.

The coverage summary will report the number of:

- SKU-store series;
- unique products;
- categories;
- departments;
- stores;
- states.

These measures provide context for interpreting the commercial and inventory KPIs presented in the remaining analysis.

Coverage KPIs calculated successfully.

SKU-store series : 300

Unique products : 30

✔ Product and business coverage validation passed.

	Coverage Measure	Value	Business Meaning
0	SKU-Store Series	300	Product-store combinations included in the ana...
1	Unique Products	30	Distinct products represented in the scope
2	Categories	3	Major product groups represented
3	Departments	3	Detailed product divisions represented
4	Stores	10	Retail locations represented
5	States	3	Geographic markets represented

PRODUCT AND BUSINESS COVERAGE SUMMARY

=====

SKU-store series : 300

Unique products : 30

Categories : 3

Departments : 3

Stores : 10

States : 3

✔ Section 6.3 completed: Product and business coverage confirmed.

Product and Business Coverage Summary

The approved development scope contains **300 SKU-store series**, representing **30 unique products** distributed across **3 categories**, **3 departments**, **10 stores**, and **3 states**.

Each SKU-store series maps consistently to one product hierarchy and geographic profile, confirming that the scoped dataset is suitable for category, department, store, and state-level analysis.

Coverage Measure	Result
SKU-Store Series	300
Unique Products	30
Categories	3
Departments	3

Coverage Measure	Result
Stores	10
States	3

✔ Section 6.3 is complete.

6.4 Simulated Inventory KPIs

This subsection provides a high-level view of the simulated inventory position associated with the approved development scope.

The inventory overview will summarise:

- average inventory on hand per SKU-store-date record;
- average inventory on order per SKU-store-date record;
- estimated inventory value at the latest available snapshot;
- historical low-stock observations;
- historical excess-stock observations;
- historical stockout observations, where available.

Important: Inventory records were simulated in Notebook 01 using documented assumptions. These KPIs support analytical development and must not be interpreted as observed retailer inventory values.

Simulated inventory KPIs calculated successfully.

Latest snapshot date : 2016-04-24

Latest snapshot rows : 300

Inventory statuses : 4

✔ Simulated inventory KPI validation passed.

	KPI	Value	Unit	Business Meaning
0	Average Inventory on Hand	19.00	Units per SKU-Store-Date	Average simulated closing stock across histori...
1	Average Inventory on Order	4.81	Units per SKU-Store-Date	Average simulated units awaiting receipt
2	Latest Estimated Inventory Value	14,170.55	Currency Units	Estimated inventory value at the latest snapshot
3	Low-Stock Observations	67,528	Historical Records	Historical records classified as Low Stock
4	Excess-Stock Observations	223,381	Historical Records	Historical records classified as Overstock
5	Stockout Observations	15,318	Historical Records	Historical records classified as Stockout

SIMULATED INVENTORY KPI SUMMARY

```
=====
Latest snapshot date      : 2016-04-24
Average inventory on hand : 19.00
Average inventory on order : 4.81
Latest inventory value    : 14,170.55
Low-stock observations    : 67,528
Excess-stock observations : 223,381
Stockout observations     : 15,318
```

✅ Section 6.4 completed: Simulated inventory KPIs calculated.

Simulated Inventory KPI Summary

The simulated inventory dataset provides a high-level view of historical stock conditions across the approved 300 SKU-store development scope.

Inventory KPI	Result
Average Inventory on Hand	19.00 units per SKU-store-date
Average Inventory on Order	4.81 units per SKU-store-date
Latest Estimated Inventory Value	14,170.55 currency units
Low-Stock Observations	67,528
Excess-Stock Observations	223,381
Stockout Observations	15,318

The latest inventory snapshot contains all **300 SKU-store series** as of **24 April 2016**, with an estimated total inventory value of **14,170.55 currency units**.

Across the complete historical dataset, **Overstock** was the most frequent inventory-risk condition, followed by **Low Stock** and **Stockout** observations. These counts represent historical SKU-store-date records rather than unique products, stores, or incidents.

Important: All inventory measures are simulated and assumption-based. They support analytical development and should not be interpreted as observed retailer inventory or audited financial values.

✅ Section 6.4 is complete.

6.5 Executive KPI Summary

This subsection will consolidate the most important historical coverage, commercial performance, business coverage, and simulated inventory measures into one executive-level KPI table.

The resulting summary will support stakeholder reporting and later dashboard development.

The final table will be exported as:

reports/tables/eda_kpi_summary.csv

Executive KPI records consolidated successfully.

Total KPIs : 24

KPI groups : 4

✓ Executive KPI summary validation passed.

✓ Executive KPI summary export verification passed.

Exported KPIs : 24

Export path : C:\Users\hp\Project FORESIGHT\reports\tables\eda_kpi_summary.csv

	KPI Group	Number of KPIs
0	Business Coverage	6
1	Commercial Performance	6
2	Historical Coverage	6
3	Simulated Inventory	6

✓ Section 6.5 completed: Executive KPI summary created, validated, and exported.

Total executive KPIs: 24

Executive KPI Summary

The executive KPI table consolidates **24 headline measures** across four analytical areas:

KPI Group	Number of KPIs
Historical Coverage	6
Commercial Performance	6
Business Coverage	6
Simulated Inventory	6
Total	24

The summary combines historical coverage, sales and revenue performance, product and organisational coverage, and simulated inventory conditions into one Commercial Performance | 6 | | Business Coverage | 6 | | Simulated Inventory | 6 | | ** stakeholder-ready table.

The final KPI table was successfully exported to:

reports/tables/eda_kpi_summary.csv

This file can be reused in the final EDA memo, executive reporting, and later dashboard development.

Important: Revenue is reported in generic currency units, while margin and inventory measures are assumption-based and should not be interpreted as audited accounting or observed retailer values.

✓ Section 6.5 is complete.

✓ Section 6: Executive KPI Overview is complete.



7. Demand Trends and Seasonality

This section examines how customer demand changes over time and identifies recurring calendar patterns within the approved development scope.

Understanding trend and seasonality is essential for selecting suitable forecasting features, validation periods, and baseline models in later notebooks.

Business Question

How has demand changed over time, and when does demand usually increase or decrease?

Objectives

- analyse daily, weekly, monthly, and yearly demand trends;
 - identify major peaks, declines, and turning points;
 - compare demand across days of the week;
 - evaluate weekday and weekend behaviour;
 - examine monthly, quarterly, and seasonal patterns;
 - translate recurring patterns into forecasting and inventory-planning implications.
-

7.1 Daily Demand Trend

Daily units sold will be aggregated across the development scope to identify:

- short-term demand fluctuations;
- unusually high or low demand days;
- visible trend changes;
- potential event-related spikes.

Daily variation will be interpreted carefully because individual-day movements may contain substantial noise.

Daily demand trend data prepared successfully.

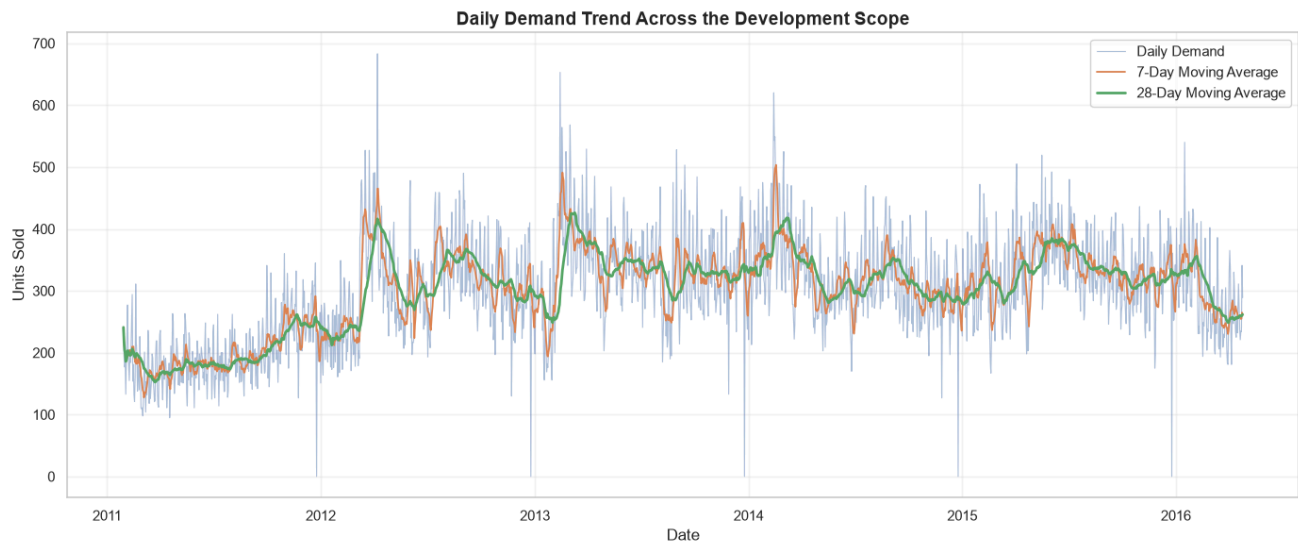
Daily observations : 1,913

Analysis period : 2011-01-29 to 2016-04-24

Total units sold : 573,932

✅ Daily demand trend validation passed.

Demand range : 0 to 683 units per day



✓ Daily demand trend figure created successfully.

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_1_daily_demand_trend.png

Daily Demand Trend Summary

Daily demand fluctuates substantially across the historical period. The **7-day moving average** highlights short-term movement, while the **28-day moving average** shows the broader demand direction more clearly.

Demand increased from the lower levels observed in 2011 and remained generally higher from 2012 onward, although several temporary spikes and sharp declines occurred. Isolated zero-demand days should be investigated separately before being treated as normal demand behaviour.

The high daily variability indicates that weekly aggregation and calendar-based features may provide a more stable foundation for forecasting and inventory planning than individual daily observations.

The dataset ends on **24 April 2016**, so the apparent decline in 2016 represents only a partial year.

✓ Section 7.1 is complete.

7.2 Weekly Demand Trend

Daily demand will be aggregated to weekly level to provide a clearer view of:

- medium-term demand direction;
- recurring weekly behaviour;
- sustained growth or decline;
- periods of unusual demand.

Weekly analysis is especially important because Project FORESIGHT will produce weekly demand forecasts.

Weekly demand trend data prepared successfully.

Weekly observations : 274

Analysis period : 2011-01-29 to 2016-04-24

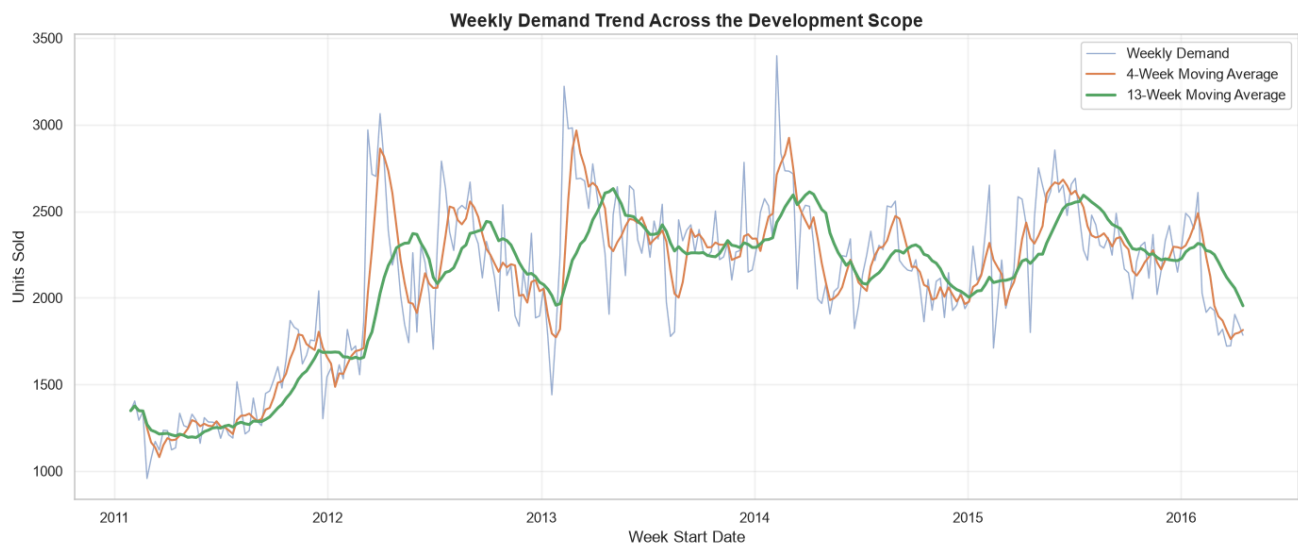
Total units sold : 573,932

✅ Weekly demand trend validation passed.

Complete weeks : 273

Partial weeks : 1

Weekly demand range : 630 to 3,397 units



✅ Weekly demand trend figure created successfully.

Complete weeks plotted : 273

Partial weeks excluded : 1

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_2_weekly_demand_trend.png

Weekly Demand Trend Summary

Weekly aggregation provides a clearer view of sustained demand movement than the highly variable daily series.

Demand increased substantially from the lower weekly levels observed in 2011 and remained generally higher from 2012 onward. However, the trend was not continuously upward; several recurring peaks were followed by periods of normalisation or decline.

The **4-week moving average** captures short-term movement, while the **13-week moving average** highlights the broader quarterly demand direction. These recurring changes support the use of lag, rolling-average, and seasonal calendar features in the weekly forecasting models.

The single partial week was excluded from the chart to ensure that all displayed weekly totals represent comparable seven-day periods.

The apparent decline in 2016 should be interpreted cautiously because the dataset ends on **24 April 2016**.

✅ Section 7.2 is complete.

7.3 Monthly Demand Trend

Monthly demand will be analysed to identify:

- longer-term business movement;
- strong and weak months;
- recurring monthly peaks;
- possible seasonal demand cycles.

Monthly demand trend data prepared successfully.

Monthly observations : 64

Analysis period : 2011-01-29 to 2016-04-24

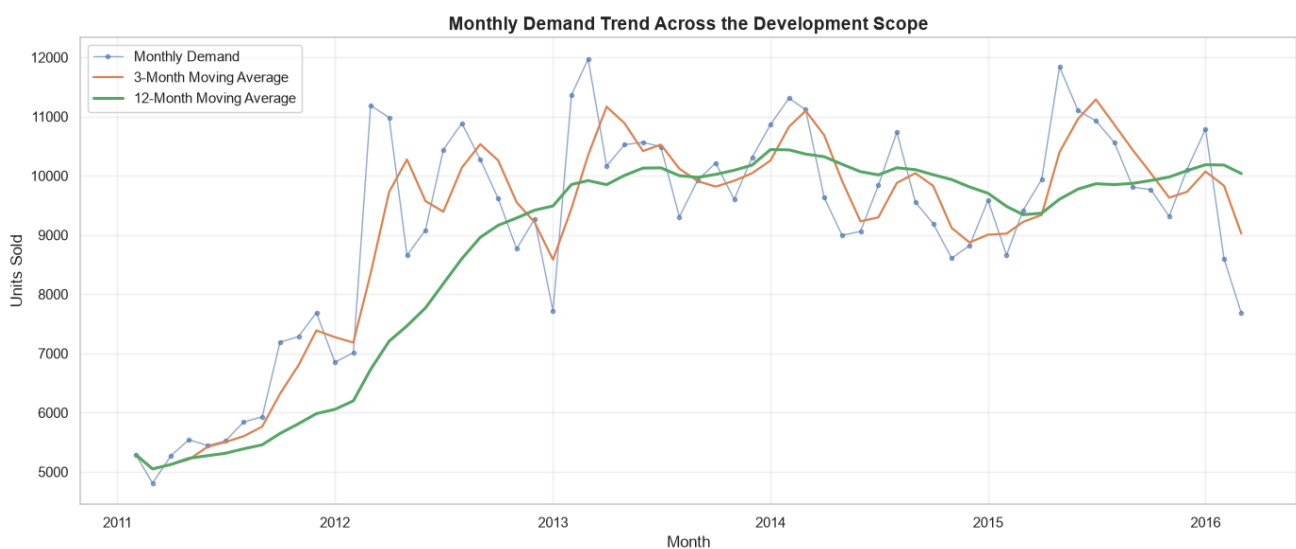
Total units sold : 573,932

✅ Monthly demand trend validation passed.

Complete months : 62

Partial months : 2

Monthly demand range : 613 to 11,972 units



✅ Monthly demand trend figure created successfully.

Complete months plotted : 62

Partial months excluded : 2

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_3_monthly_demand_trend.png

Monthly Demand Trend Summary

Monthly aggregation confirms a substantial increase in demand from 2011 into 2012. From 2012 onward, demand remained at a generally higher level but continued to experience recurring peaks and declines.

The **3-month moving average** highlights short-term changes, while the **12-month moving average** indicates that the broader demand level became relatively stable after the initial growth period. This supports the inclusion of monthly and annual seasonal features in later forecasting models.

January 2011 and April 2016 were excluded because they contain incomplete monthly coverage, ensuring that all displayed monthly totals are directly comparable.

The final decline should be interpreted cautiously because the chart includes complete months only through **March 2016**.

✓ Section 7.3 is complete.

7.4 Yearly Demand Comparison

Yearly demand totals and average demand levels will be compared across the available historical period.

The analysis will consider that incomplete years may not be directly comparable with complete years.

Yearly demand comparison prepared successfully.

Yearly observations : 6

Complete years : 4

Partial years : 2

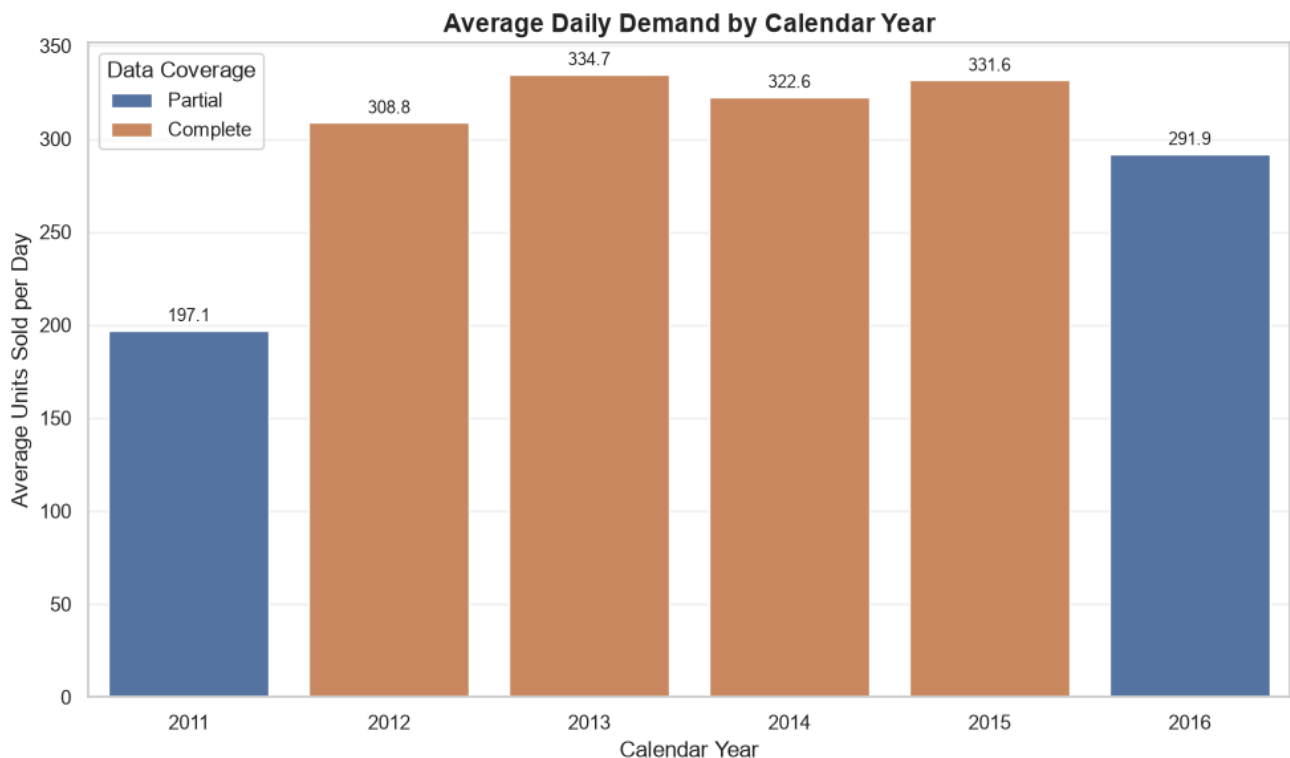
Total units sold : 573,932

	Year	Total Units Sold	Average Daily Demand	Observed Days	Coverage
0	2011	66429	197.12	337	Partial
1	2012	113018	308.79	366	Complete
2	2013	122152	334.66	365	Complete
3	2014	117735	322.56	365	Complete
4	2015	121032	331.59	365	Complete
5	2016	33566	291.88	115	Partial

✓ Yearly demand comparison validation passed.

Complete years : 4

Partial years : 2



✅ Yearly demand comparison figure created successfully.
Years displayed : 6
Complete years : 4
Partial years : 2
Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_4_yearly_average_daily_demand.png

Yearly Demand Comparison Summary

Average daily demand increased sharply from the partial 2011 period and remained at a substantially higher level during the complete years from 2012 to 2015.

Among the complete years, **2013 recorded the highest average daily demand at 334.7 units**, while 2012 recorded the lowest at 308.8 units. Demand declined moderately in 2014 and recovered in 2015, indicating fluctuation around a relatively stable higher demand level rather than continuous yearly growth.

The values for **2011 and 2016** should be interpreted cautiously because both years contain partial historical coverage.

✅ Section 7.4 is complete.

7.5 Day-of-Week Pattern

Average demand will be compared across calendar weekdays to identify whether particular days consistently experience higher or lower sales.

Day-of-week demand summary prepared successfully.

Weekdays represented : 7

Observed day range : 273 to 274

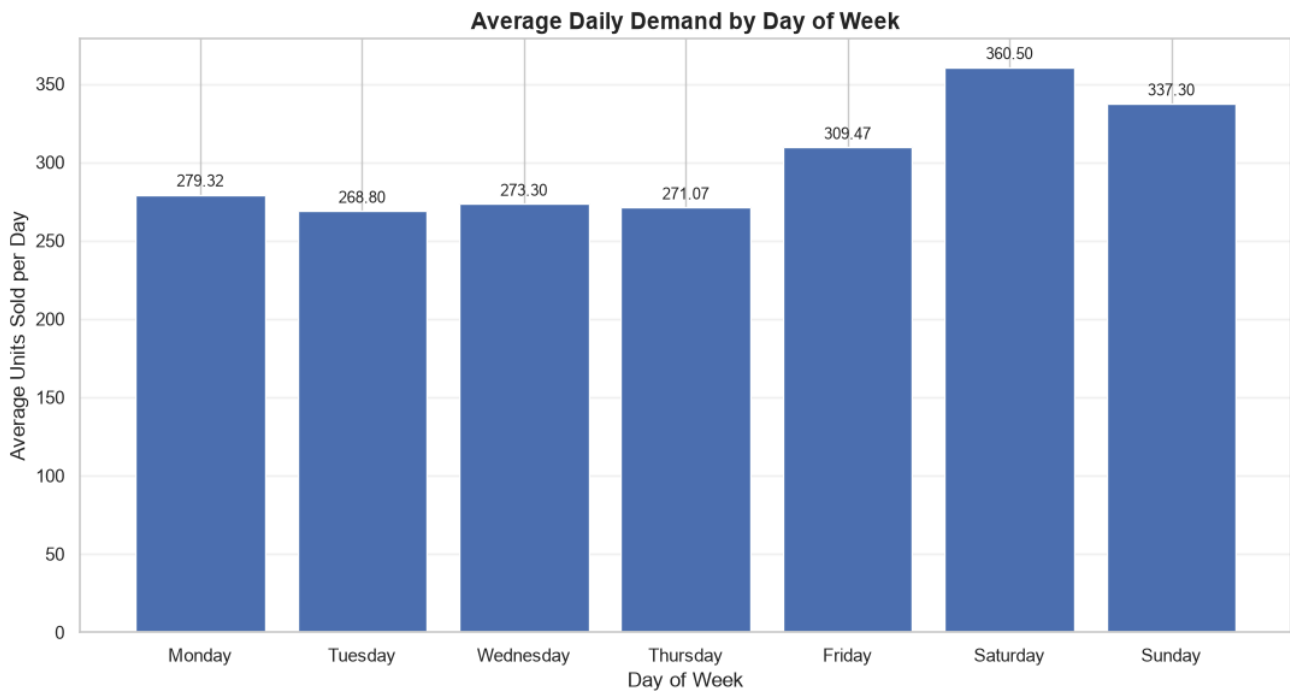
Average demand range: 268.80 to 360.50

	Day of Week	Observed Days	Total Units Sold	Average Daily Demand
0	Monday	273	76254	279.32
1	Tuesday	273	73382	268.80
2	Wednesday	273	74612	273.30
3	Thursday	273	74002	271.07
4	Friday	273	84484	309.47
5	Saturday	274	98777	360.50
6	Sunday	274	92421	337.30

✅ Day-of-week demand validation passed.

Highest-demand day : Saturday (360.50 units)

Lowest-demand day : Tuesday (268.80 units)



✅ Day-of-week demand figure created successfully.

Highest-demand day : Saturday (360.50 units)

Lowest-demand day : Tuesday (268.80 units)

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_5_day_of_week_demand_pattern.png

Day-of-Week Demand Summary

Demand follows a clear weekly pattern. Average daily demand remains relatively stable from Monday to Thursday, increases on Friday, and reaches its highest level during the weekend.

Saturday recorded the highest average demand at 360.50 units, followed by Sunday at 337.30 units. **Tuesday recorded the lowest average demand at 268.80 units.**

This pattern suggests that weekday information should be included as a forecasting feature. Inventory and replenishment planning should also account for the higher demand expected on Fridays and weekends.

✅ Section 7.5 is complete.

7.6 Weekday versus Weekend Demand

Demand during weekdays and weekends will be compared using:

- total units sold;
- average daily demand;
- revenue, where appropriate.

The comparison will use averages as well as totals because weekdays and weekends contain different numbers of observations.

Weekday and weekend demand summary prepared successfully.

Comparison groups : 2

Observed days : 1,913

Total units sold : 573,932

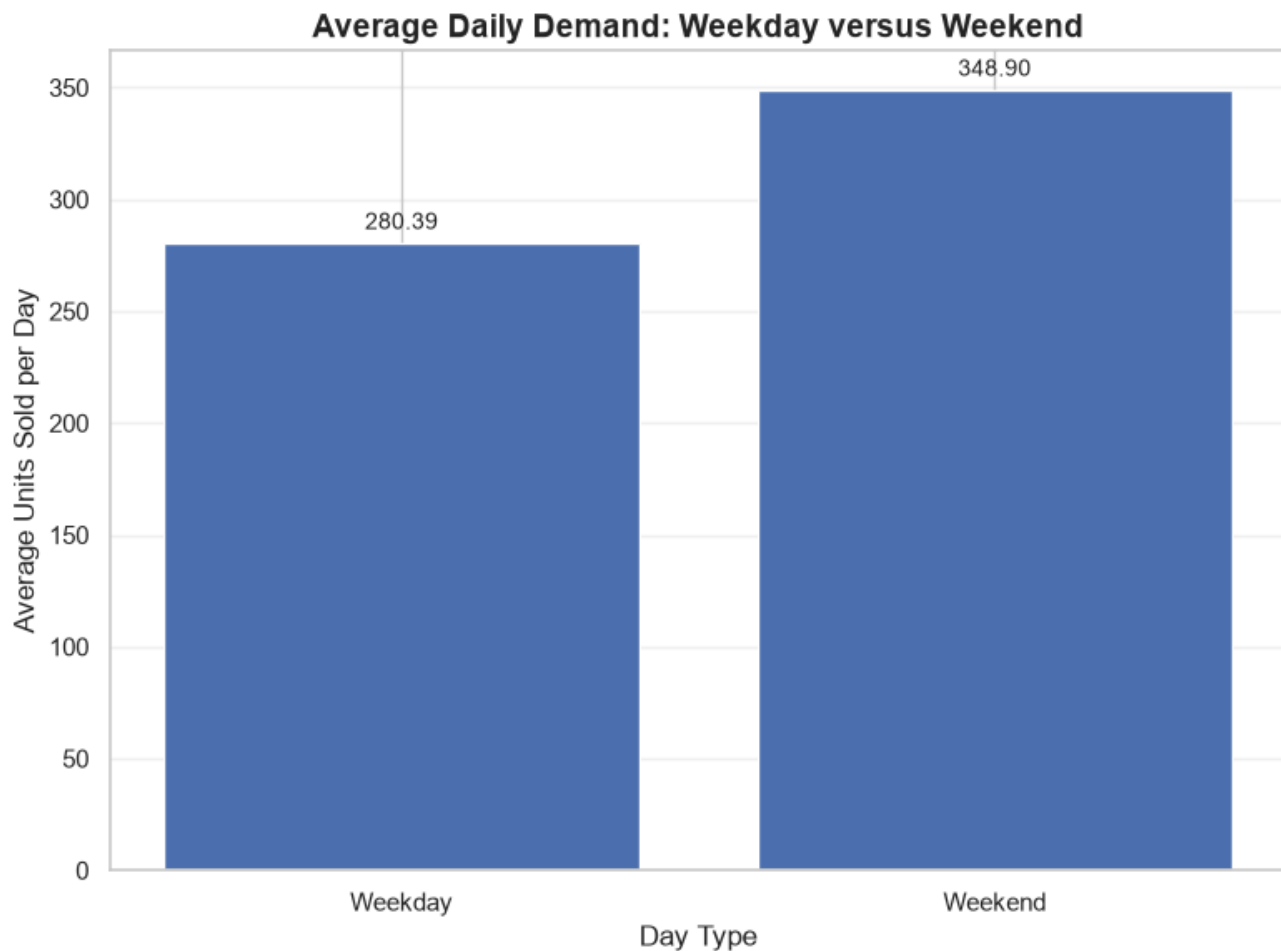
	Day Type	Observed Days	Total Units Sold	Average Daily Demand	Total Revenue	Average Daily Revenue
0	Weekday	1365	382734	280.39	1,128,420.81	826.68
1	Weekend	548	191198	348.90	591,895.78	1,080.10

✅ Weekday versus weekend validation passed.

Weekday average demand : 280.39 units

Weekend average demand : 348.90 units

Weekend demand uplift : 24.43%



✅ Weekday versus weekend figure created successfully.

Weekday average demand : 280.39 units

Weekend average demand : 348.90 units

Weekend demand uplift : 24.43%

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_6_weekday_weekend_demand.png

Weekday versus Weekend Demand Summary

Average weekend demand reached **348.90 units per day**, compared with **280.39 units per weekday**, representing a **24.43% weekend uplift**.

Although weekdays generated more total units because they contain more observed days, weekends consistently produced stronger demand on a per-day basis. Average daily revenue also increased from **826.68 currency units on weekdays** to **1,080.10 currency units on weekends**.

This pattern supports the use of a weekend indicator in forecasting models. Replenishment and staffing decisions should also account for the stronger demand expected on Saturdays and Sundays.

✔ Section 7.6 is complete.

7.7 Monthly, Quarterly, and Seasonal Patterns

Demand will be summarised by:

- month;
- quarter;
- defined calendar season.

These summaries will help identify recurring demand periods that may later support calendar and seasonality features.

Calendar-pattern summaries prepared successfully.

Months represented : 12

Quarters represented : 4

Seasons represented : 4

Total units covered : 573,932

	Month	Observed Days	Average Daily Demand
0	January	158	293.80
1	February	170	307.25
2	March	186	302.07
3	April	174	301.64
4	May	155	293.92
5	June	150	301.69
6	July	155	304.61
7	August	155	305.39
8	September	150	303.31
9	October	155	296.64
10	November	150	290.50
11	December	155	298.03

	Quarter	Observed Days	Average Daily Demand
0	Q1	514	301.24
1	Q2	479	299.16
2	Q3	460	304.45
3	Q4	460	295.10

	Season	Observed Days	Average Daily Demand
0	Winter	483	299.89
1	Spring	515	299.47
2	Summer	460	303.92
3	Autumn	455	296.82

✅ Calendar-pattern validation passed.

Highest-demand month : February (307.25 units)

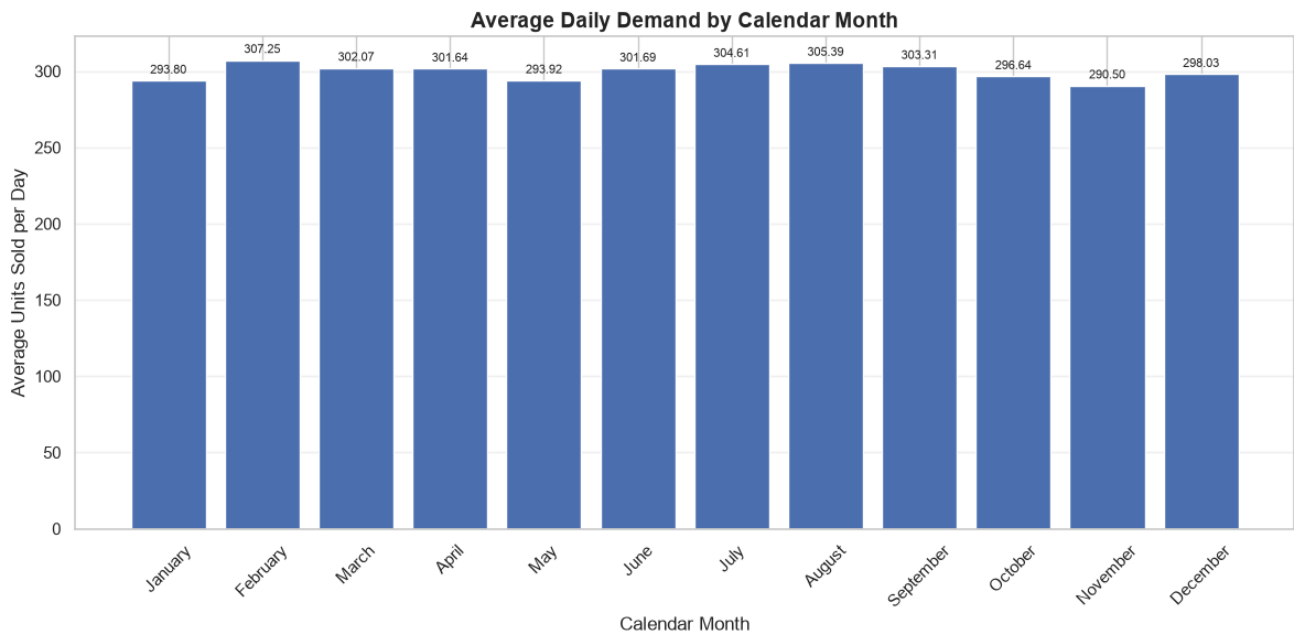
Lowest-demand month : November (290.50 units)

Highest-demand quarter : Q3 (304.45 units)

Lowest-demand quarter : Q4 (295.10 units)

Highest-demand season : Summer (303.92 units)

Lowest-demand season : Autumn (296.82 units)

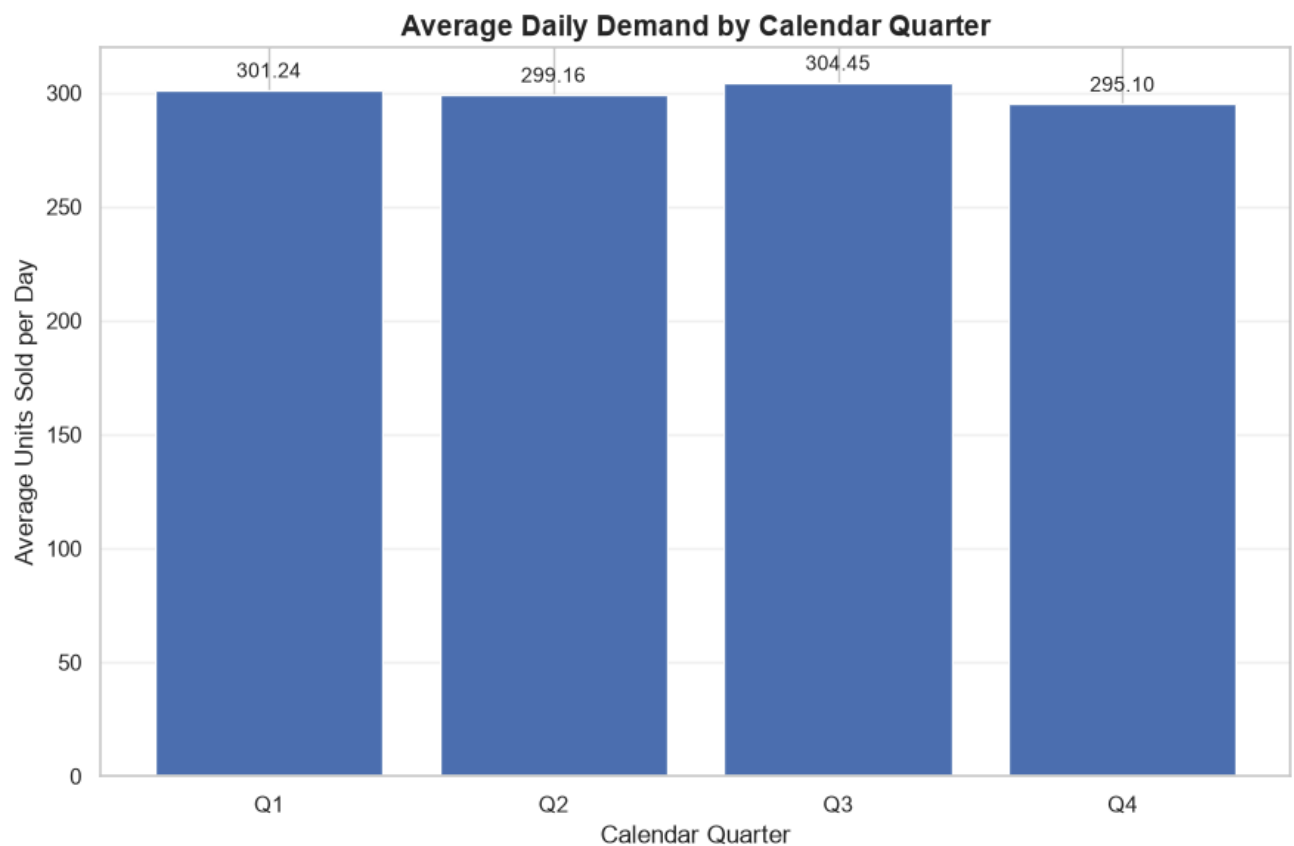


✅ Monthly demand-pattern figure created successfully.

Highest-demand month : February (307.25 units)

Lowest-demand month : November (290.50 units)

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_7_monthly_demand_p
attern.png

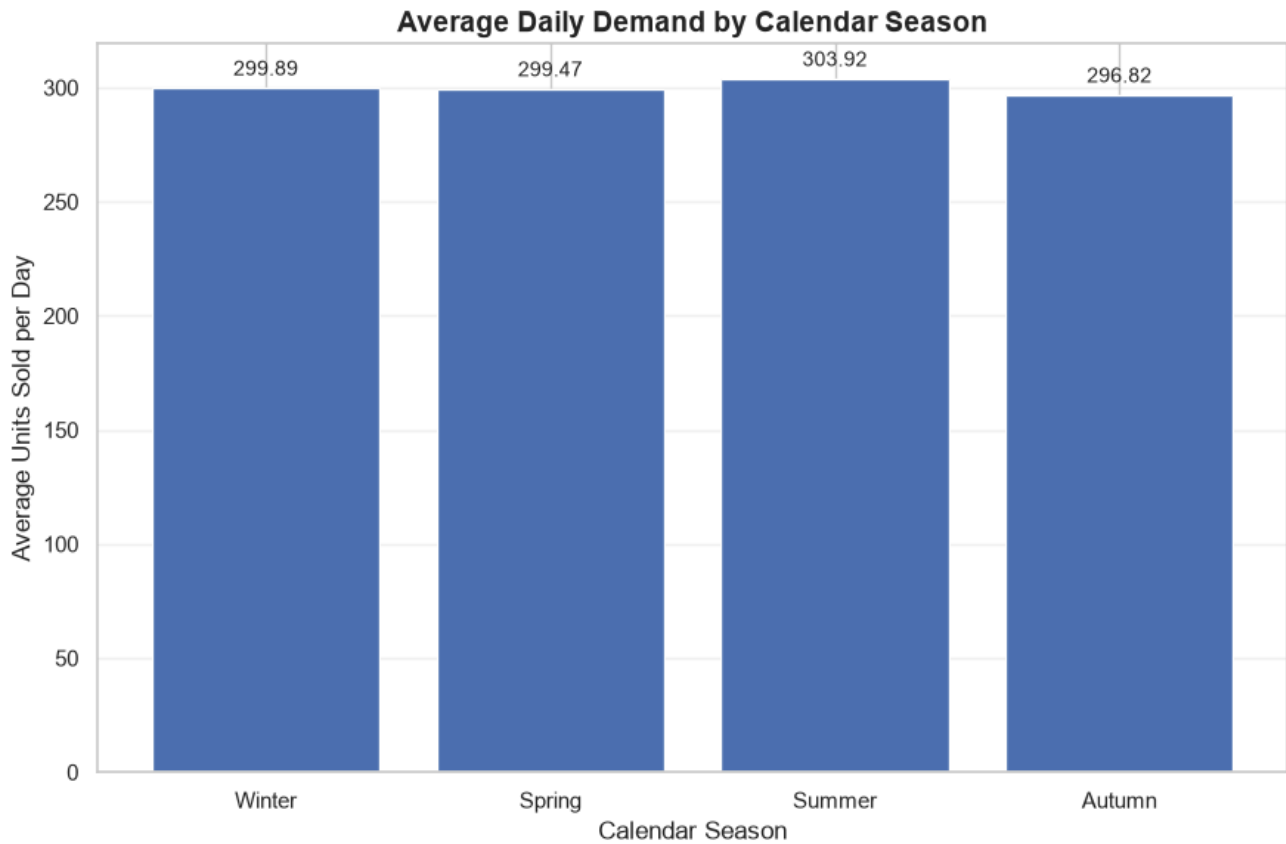


✓ Quarterly demand-pattern figure created successfully.

Highest-demand quarter : Q3 (304.45 units)

Lowest-demand quarter : Q4 (295.10 units)

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_7_quarterly_demand_pattern.png



✓ Seasonal demand-pattern figure created successfully.

Highest-demand season : Summer (303.92 units)

Lowest-demand season : Autumn (296.82 units)

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\07_7_seasonal_demand_pattern.png

Monthly, Quarterly, and Seasonal Pattern Summary

Recurring calendar patterns are present, although their differences are moderate.

February recorded the highest monthly average demand at **307.25 units per day**, while **November** recorded the lowest at **290.50 units**. At the quarterly level, **Q3** was strongest at **304.45 units per day**, while **Q4** was weakest at **295.10 units**. **Summer** recorded the highest seasonal demand at **303.92 units per day**, compared with **296.82 units during Autumn**.

These results support including month, quarter, and season features in forecasting models. However, their demand differences are smaller than the weekday and weekend effects, so they should be treated as supporting rather than dominant demand drivers.

✓ Section 7.7 is complete.

7.8 Trend and Seasonality Summary

The most important findings from this section will be documented using:

- analytical observation;
- business interpretation;
- forecasting implication;
- inventory-planning implication.

Interpretation rule: Historical patterns describe observed associations and recurring behaviour. They do not guarantee that the same pattern will continue unchanged in future periods.

Trend and seasonality findings prepared successfully.

Strongest complete year : 2013 (334.66 units/day)

Weakest complete year : 2012 (308.79 units/day)

Weekday demand gap : 91.70 units/day

Monthly demand gap : 16.75 units/day

Seasonal demand gap : 7.11 units/day

Trend and seasonality summary prepared successfully.

Findings documented : 5

Summary fields : 5

✅ Trend and seasonality summary validation passed.

Validated findings : 5

Validated fields : 5

✅ Section 7.8 completed: Trend and seasonality findings consolidated.

Trend and Seasonality Summary

Demand increased from the lower levels observed in 2011 and remained generally higher from 2012 onward, although no continuous yearly growth pattern was observed.

The strongest recurring effect was the weekly cycle: weekend demand was **24.43% higher** than weekday demand, with Saturday recording the highest average demand. Monthly, quarterly, and seasonal differences were present but comparatively moderate.

These findings support using trend, lag, rolling-average, weekday, weekend, month, quarter, and season features in later forecasting models. Inventory planning should place greater emphasis on recent demand and weekend peaks than on broad seasonal adjustments.

Historical patterns describe observed associations and may not continue unchanged in future periods.

✅ Section 7.8 is complete.

✅ Section 7: Demand Trends and Seasonality is complete.



8. Business Performance Analysis

This section compares business performance across product hierarchies, stores, states, products, and SKU-store combinations.

The purpose is to identify which parts of the business generate the strongest demand and revenue, and where weaker performance may require further investigation.

Business Question

Which categories, departments, stores, states, products, and SKU-store combinations perform best or worst?

Objectives

- compare demand and revenue across categories and departments;
- evaluate store and state performance;
- measure product and SKU-store contribution;
- identify consistently strong and weak business segments;
- distinguish high-demand performance from high-revenue performance;
- prepare business-level summary tables for reporting and dashboards.

8.1 Category Performance

Categories will be compared using:

- total units sold;
- total revenue;
- average daily demand;
- revenue contribution;
- estimated gross margin, where available.

This analysis will show which broad product groups contribute most to the business.

Category performance summary prepared successfully.

Categories analysed : 3

Total units covered : 573,932

Total revenue covered: 1,720,316.59

	Category	Active SKU Series	Total Units Sold	Average Daily Demand	Demand Contribution (%)	Total Revenue	Revenue Contribution (%)	Estimated Gross Margin	Margin Contribution (%)	Demand Rank	Revenue Rank
0	FOODS	100	235002	122.84	40.95	586,096.99	34.07	189,357.70	34.91	1	2
1	HOBBIES	100	195456	102.17	34.06	417,160.26	24.25	129,560.98	23.89	2	3
2	HOUSEHOLD	100	143474	75.00	25.00	717,059.34	41.68	223,510.50	41.21	3	1

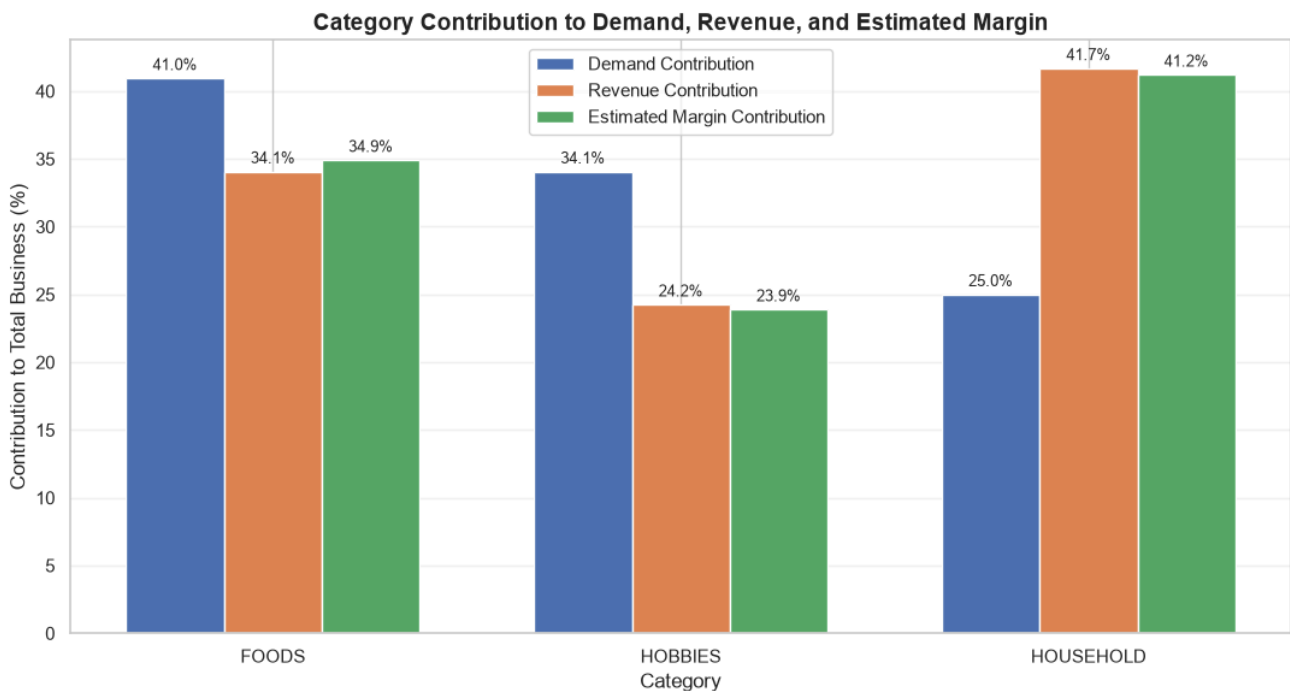
✅ Category performance validation passed.

Highest-demand category : FOODS (235,002 units)

Lowest-demand category : HOUSEHOLD (143,474 units)

Highest-revenue category: HOUSEHOLD (717,059.34)

Lowest-revenue category : HOBBIES (417,160.26)



✅ Category contribution figure created successfully.

Demand leader : FOODS

Revenue leader : HOUSEHOLD

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_1_category_contribution_comparison.png

Category Performance Summary

- **Demand leader:** FOODS contributed **41.0% of total units sold** but only **34.1% of total revenue**.
- **Revenue leader:** HOUSEHOLD contributed just **25.0% of demand** but generated the highest revenue contribution at **41.7%**.
- **Lower-value category:** HOBBIES contributed **34.1% of demand** but only **24.2% of revenue**.
- **Key insight:** High sales volume does not necessarily produce the highest revenue or margin.
- **Margin pattern:** Estimated-margin contribution broadly follows revenue, with **HOUSEHOLD first, FOODS second, and HOBBIES third**.
- **Business implication:** Use FOODS to support volume, protect HOUSEHOLD as the strongest financial contributor, and investigate HOBBIES for **pricing, product mix, and margin improvement**.

Estimated margins are assumption-based because unit costs were estimated rather than obtained from accounting records.

✅ Section 8.1: Category Performance is complete.

8.2 Department Performance

Departments will be evaluated using demand, revenue, and contribution measures.

Department-level analysis provides a more detailed view than category-level analysis and helps identify strong or weak product groups within each category.

Department performance summary prepared successfully.

Departments analysed : 3

Total units covered : 573,932

Total revenue covered: 1,720,316.59

	Depart ment	Active SKU Series	Total Units Sold	Average Daily Demand	Demand Contributi on (%)	Total Reven ue	Average Revenue per Unit	Revenue Contributi on (%)	Estimated Gross Margin	Margin Contributi on (%)	Dema nd Rank	Reven ue Rank
0	FOOD S_1	100	235002	122.84	40.95	586,096.99	2.49	34.07	189,357.70	34.91	1	2
1	HOBBI ES_1	100	195456	102.17	34.06	417,160.26	2.13	24.25	129,560.98	23.89	2	3
2	HOUS EHOLD _1	100	143474	75.00	25.00	717,059.34	5.00	41.68	223,510.50	41.21	3	1

✅ Department performance validation passed.

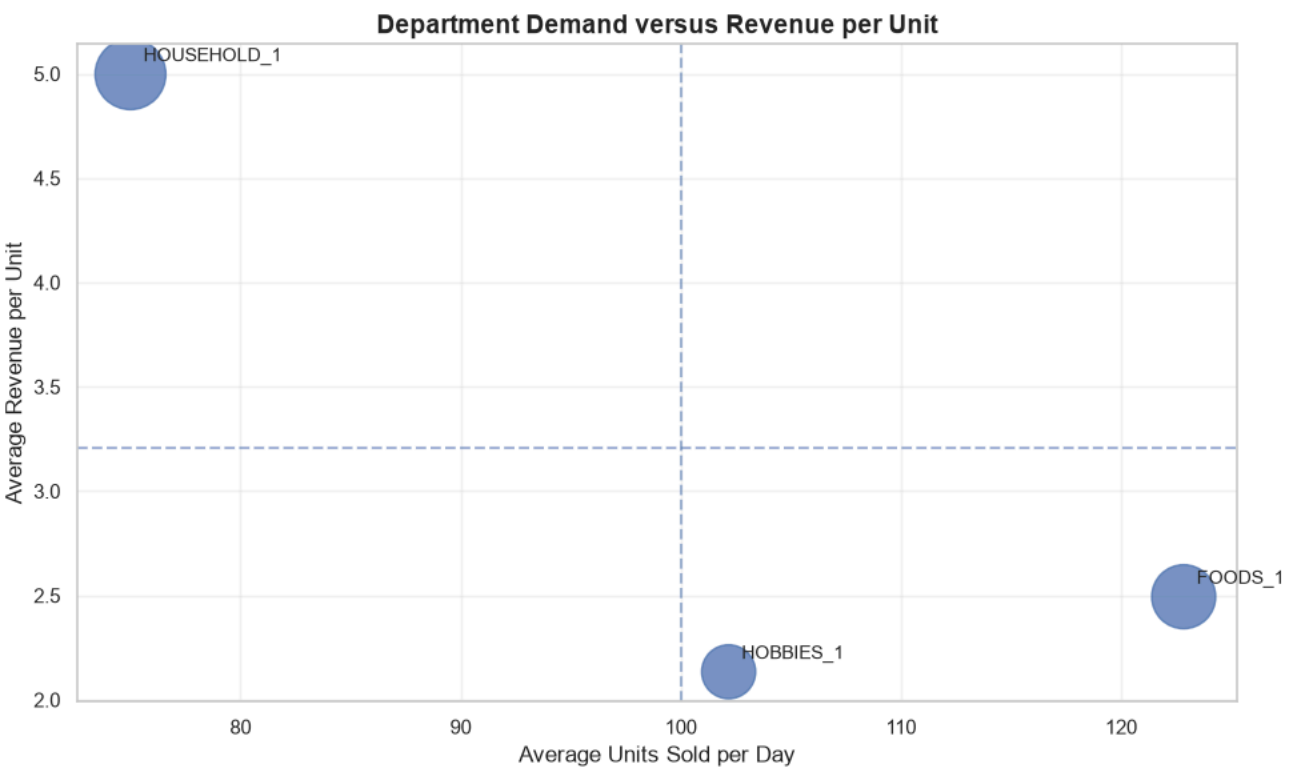
Highest-demand department : FOODS_1 (235,002 units)

Lowest-demand department : HOUSEHOLD_1 (143,474 units)

Highest-revenue department: HOUSEHOLD_1 (717,059.34)

Lowest-revenue department : HOBBIES_1 (417,160.26)

Highest revenue per unit : HOUSEHOLD_1 (5.00)



✅ Department demand-value figure created successfully.

Demand leader : FOODS_1

Unit-value leader : HOUSEHOLD_1

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_2_department_demand_value_comparison.png

Department Performance Summary

- **Demand leader:** FOODS_1 contributed **40.95% of total units sold** and **34.07% of total revenue**.
- **Revenue leader:** HOUSEHOLD_1 contributed only **25.00% of demand** but generated the highest revenue contribution at **41.68%**.
- **Highest unit value:** HOUSEHOLD_1 earned **5.00 revenue units per item**, compared with **2.49** for FOODS_1 and **2.13** for HOBBIES_1.
- **Weakest financial conversion:** HOBBIES_1 contributed **34.06% of demand** but only **24.25% of revenue**.
- **Margin pattern:** Estimated-margin rankings broadly follow revenue rankings, with **HOUSEHOLD_1 contributing the most estimated margin**.
- **Scope note:** Each category currently contains only one department, so department findings closely mirror category findings.
- **Business implication:** Prioritise FOODS_1 for volume, protect HOUSEHOLD_1 for financial value, and review HOBBIES_1 for **pricing, product mix, and margin improvement**.

Estimated margins are assumption-based because actual product-cost data was unavailable.

✅ Section 8.2: Department Performance is complete.

8.3 Store Performance

Stores will be compared using:

- total units sold;
- total revenue;
- average daily demand;
- number of active SKU series;
- contribution to overall business performance.

The analysis will help identify stores with unusually strong or weak demand patterns.

Store performance summary prepared successfully.

Stores analysed : 10

States represented : 3

Total units covered : 573,932

Total revenue covered : 1,720,316.59

	Store	State	Active SKU Series	Total Units Sold	Average Daily Demand	Demand Contribution (%)	Total Revenue	Average Revenue per Unit	Revenue Contribution (%)	Estimated Gross Margin	Margin Contribution (%)	Demand Rank	Revenue Rank
0	CA_3	CA	30	96898	50.65	16.88	282,961.17	2.92	16.45	87,779.19	16.18	1	1
1	CA_2	CA	30	69948	36.56	12.19	238,962.37	3.42	13.89	75,296.93	13.88	2	2
2	CA_1	CA	30	69797	36.49	12.16	198,176.11	2.84	11.52	62,399.34	11.50	3	3
3	WI_1	WI	30	59923	31.32	10.44	181,013.98	3.02	10.52	57,529.74	10.61	4	4
4	TX_2	TX	30	56753	29.67	9.89	167,006.06	2.94	9.71	53,037.94	9.78	5	5
5	WI_2	WI	30	50987	26.65	8.88	141,120.32	2.77	8.20	44,523.03	8.21	6	7
6	WI_3	WI	30	47762	24.97	8.32	144,361.95	3.02	8.39	45,422.80	8.37	7	6
7	TX_3	TX	30	43846	22.92	7.64	135,255.43	3.08	7.86	43,140.12	7.95	8	8
8	TX_1	TX	30	42275	22.10	7.37	120,268.34	2.84	6.99	38,118.11	7.03	9	9
9	CA_4	CA	30	35743	18.68	6.23	111,190.86	3.11	6.46	35,181.98	6.49	10	10

✔ Store performance validation passed.

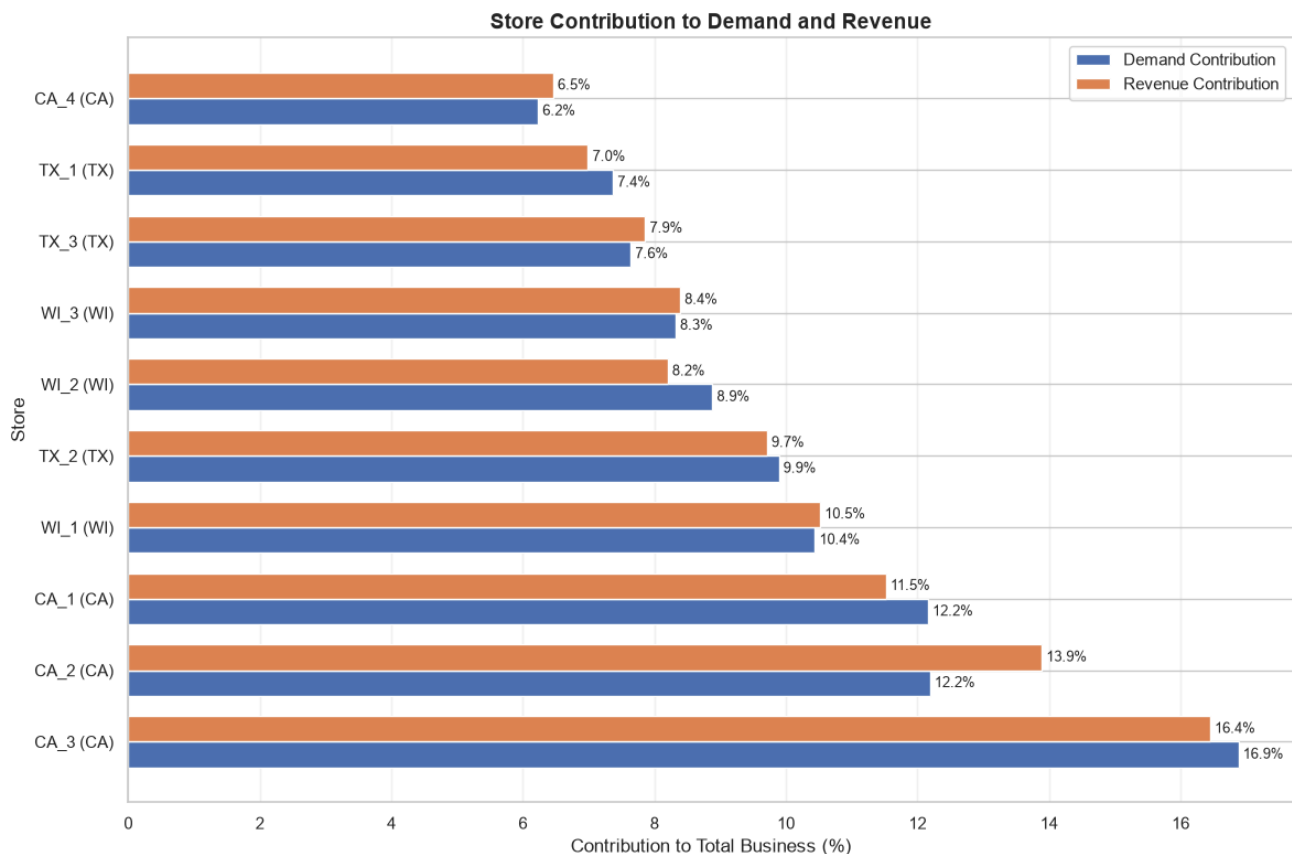
Highest-demand store : CA_3 (96,898 units)

Lowest-demand store : CA_4 (35,743 units)

Highest-revenue store: CA_3 (282,961.17)

Lowest-revenue store : CA_4 (111,190.86)

Highest revenue/unit : CA_2 (3.42)



✅ Store contribution figure created successfully.

Demand leader : CA_3

Revenue leader : CA_3

Revenue/unit leader: CA_2

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_3_store_contribution_comparison.png

Store Performance Summary

- **Demand leader:** CA_3 recorded the highest demand with **96,898 units**, contributing **16.88%** of total units sold.
- **Revenue leader:** CA_3 also generated the highest revenue at **282,961.17**, contributing **16.45%** of total revenue.
- **Highest unit value:** CA_2 recorded the highest average revenue per unit at **3.42**, which explains why its revenue contribution exceeded its demand contribution.
- **Lowest-performing store:** CA_4 recorded the lowest demand at **35,743 units** and the lowest revenue at **111,190.86**.
- **SKU coverage:** Every store contained **30 active SKU series**, so store-level performance differences were not caused by unequal product coverage.
- **Geographic pattern:** California stores occupied the top three positions for demand and revenue, suggesting stronger overall performance in California.
- **Ranking consistency:** Demand and revenue rankings were generally similar, although WI_2 and WI_3 changed positions because of differences in revenue generated per unit.

- **Business implication:** CA_3 should be treated as a high-priority store for stock availability and replenishment. CA_4 requires further investigation into demand, local market conditions, pricing, and product-level performance.

Estimated margin findings remain assumption-based because actual unit-cost data was unavailable.

✔ Section 8.3: Store Performance is complete.

8.4 State Performance

State-level demand and revenue will be compared across the available geographic regions.

Differences will be interpreted alongside the number of stores and SKU-store series represented in each state.

State performance summary prepared successfully.

States analysed : 3
Stores covered : 10
SKU series covered : 300
Total units covered : 573,932
Total revenue covered : 1,720,316.59

	State	Store Count	Active SKU Series	Total Units Sold	Average Daily Demand	Average Daily Demand per Store	Demand Contribution (%)	Total Revenue	Average Revenue per Unit	Revenue Contribution (%)	Estimated Gross Margin	Margin Contribution (%)	Demand Rank	Revenue Rank
0	CA	4	120	272386	142.39	35.60	47.46	831,290.51	3.05	48.32	260,657.44	48.05	1	1
1	WI	3	90	158672	82.94	27.65	27.65	466,496.25	2.94	27.12	147,475.57	27.19	2	2
2	TX	3	90	142874	74.69	24.90	24.89	422,529.83	2.96	24.56	134,296.17	24.76	3	3

✔ State performance validation passed.

Highest-demand state : CA (272,386 units)
Lowest-demand state : TX (142,874 units)
Highest-revenue state : CA (831,290.51)
Lowest-revenue state : TX (422,529.83)
Highest demand per store : CA (35.60 units/day/store)

State Performance Comparison



✅ State performance figure created successfully.

Total-demand leader : CA

Revenue leader : CA

Demand-per-store leader: CA

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_4_state_performance_comparison.png

State Performance Summary

- **Overall leader:** California contributed **47.46% of total demand** and **48.32% of total revenue**.
- **Highest store productivity:** California also recorded the highest average daily demand per store at **35.60 units**.
- **Second position:** Wisconsin contributed **27.65% of demand** and **27.12% of revenue**, with **27.65 units per day per store**.
- **Lowest performance:** Texas contributed **24.89% of demand** and **24.56% of revenue**, with the lowest store-level demand at **24.90 units per day**.
- **Revenue efficiency:** Average revenue per unit was similar across states, ranging from **2.94 to 3.05**, so demand volume is the main driver of state-level revenue differences.
- **Key insight:** California's leadership is not explained only by having more stores; its stores also perform better individually.
- **Business implication:** Maintain strong stock availability in California, investigate successful CA store practices, and review demand-generation opportunities in Texas.

Estimated margin values remain assumption-based because actual unit-cost data was unavailable.

✅ Section 8.4: State Performance is complete.

8.5 Product Performance

Individual products will be evaluated using:

- total demand;
- total revenue;
- average selling price;
- active selling days;
- average daily demand;
- estimated financial contribution.

This analysis will identify products that contribute strongly or weakly to overall performance.

Product performance summary prepared successfully.

Product field used : item_id
Products analysed : 30
Stores represented : 10
Total units covered : 573,932
Total revenue covered : 1,720,316.59

✅ Product selling-day activity corrected successfully.

Minimum selling-day frequency : 31.21%

Maximum selling-day frequency : 99.48%

	Product	Category	Department	Store Count	Total Units Sold	Average Daily Demand	Active Selling Days	Selling-Day Frequency (%)	Total Revenue	Average Selling Price	Revenue Contribution (%)	Estimated Gross Margin	Margin Contribution (%)	Demand Rank	Revenue Rank
0	FOODS_1_004	FOODS	FOODS_1	10	129481	67.68	1366	71.41	235,998.02	1.82	13.72	76,393.79	14.08	1	1
1	HOBBIES_1_008	HOBBIES	HOBBIES_1	10	87645	45.82	1711	89.44	41,573.46	0.47	2.42	13,146.75	2.42	2	15
2	HOBBIES_1_004	HOBBIES	HOBBIES_1	10	39175	20.48	1900	99.32	176,967.84	4.52	10.29	54,453.25	10.04	3	3
3	HOUSEHOLD_1_007	HOUSEHOLD	HOUSEHOLD_1	10	36078	18.86	1700	88.87	54,163.44	1.50	3.15	16,500.60	3.04	4	10
4	HOUSEHOLD_1_004	HOUSEHOLD	HOUSEHOLD_1	10	31060	16.24	1719	89.86	69,724.06	2.24	4.05	22,052.60	4.07	5	7
5	FOODS_1_005	FOODS	FOODS_1	10	22914	11.98	1467	76.69	76,448.78	3.34	4.44	24,288.84	4.48	6	6
6	FOODS_1_006	FOODS	FOODS_1	10	22320	11.67	1903	99.48	49,012.57	2.20	2.85	16,516.80	3.04	7	12
7	HOUSEHOLD_1_005	HOUSEHOLD	HOUSEHOLD_1	10	18066	9.44	1658	86.67	181,181.03	10.03	10.53	57,343.14	10.57	8	2
8	HOBBIES_1_009	HOBBIES	HOBBIES_1	10	15657	8.18	1871	97.80	27,002.52	1.72	1.57	8,181.81	1.51	9	21
9	HOUSEHOLD_1_006	HOUSEHOLD	HOUSEHOLD_1	10	14632	7.65	1588	83.01	101,747.92	6.95	5.91	30,580.88	5.64	10	4

✅ Product performance validation passed.

Highest-demand product : FOODS_1_004 (129,481 units)

Lowest-demand product : HOBBIES_1_003 (1,435 units)

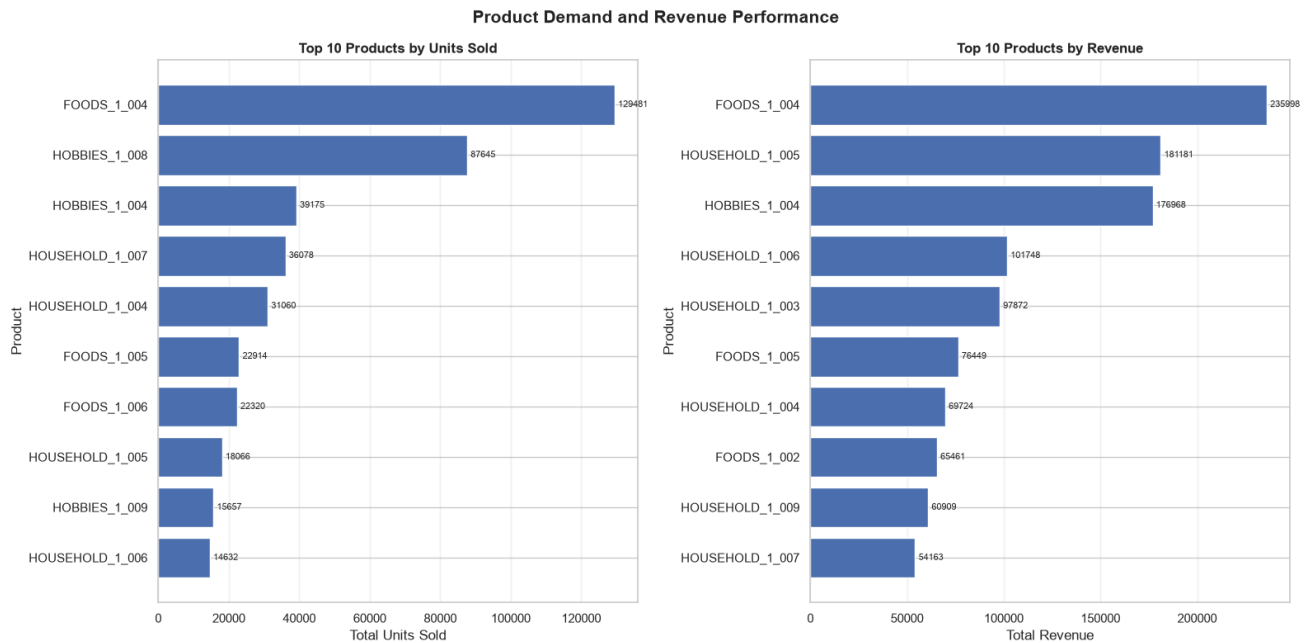
Highest-revenue product : FOODS_1_004 (235,998.02)

Lowest-revenue product : HOBBIES_1_003 (4,252.89)

Highest-priced product : HOUSEHOLD_1_009 (17.97)

Highest selling frequency: FOODS_1_006 (99.48%)

Lowest selling frequency : HOBBIES_1_003 (31.21%)



✅ Product performance figure created successfully.

Demand leader : FOODS_1_004

Revenue leader : FOODS_1_004

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_5_top_product_performance.png

Product Performance Summary

- **Overall leader:** FOODS_1_004 ranked first in both demand and revenue, with **129,481 units sold** and **235,998.02 in revenue**.
- **Demand concentration:** FOODS_1_004 alone contributed approximately **22.6% of total units sold**, indicating substantial dependence on one product.
- **Volume–value difference:** HOBBIES_1_008 ranked second in demand with **87,645 units**, but its low average selling price of **0.47** limited its financial contribution.
- **High-value product:** HOUSEHOLD_1_005 ranked only eighth in demand but second in revenue because of its higher average selling price of **10.03**.
- **Highest price:** HOUSEHOLD_1_009 recorded the highest average selling price at **17.97**.
- **Most consistent seller:** FOODS_1_006 recorded demand on **99.48% of observed days**.
- **Weakest product:** HOBBIES_1_003 recorded the lowest demand, revenue, and selling frequency, with only **1,435 units sold** and activity on **31.21% of days**.
- **Business implication:** Protect availability for FOODS_1_004, distinguish high-volume products from high-value products, and investigate HOBBIES_1_003 for possible repositioning, reduced inventory, or discontinuation.

Estimated margin results remain assumption-based because actual product-cost data was unavailable.

✅ Section 8.5: Product Performance is complete.

8.6 SKU-Store Performance

Performance will also be examined at the SKU-store level, which is the core analytical grain used throughout Project FORESIGHT.

This helps distinguish whether a product performs consistently across stores or depends heavily on particular locations.

SKU-store performance summary prepared successfully.

State field used : state_id

SKU-store series analysed: 300

Products represented : 30

Stores represented : 10

Total units covered : 573,932

Total revenue covered : 1,720,316.59

	SKU-Store Series	Product	Store	State	Category	Total Units Sold	Average Daily Demand	Active Selling Days	Selling-Day Frequency (%)	Total Revenue	Average Selling Price	Revenue Contribution (%)	Estimated Gross Margin	Margin Contribution (%)	Demand Rank	Revenue Rank
0	FOODS_1_004_WI_2	FOODS_DS_1_004	WI_2	WI	FOODS	18709	9.78	1255	65.60	34,075.34	1.82	1.98	11,038.31	2.03	1	2
1	FOODS_1_004_CA_3	FOODS_DS_1_004	CA_3	CA	FOODS	17508	9.15	1222	63.88	31,910.80	1.82	1.85	10,329.72	1.90	2	4
2	HOBBI_1_008_CA_3	HOBBI_DS_1_008	CA_3	CA	HOBBIES	16671	8.71	1339	69.99	7,946.68	0.48	0.46	2,500.65	0.46	3	54
3	FOODS_1_004_CA_1	FOODS_DS_1_004	CA_1	CA	FOODS	15797	8.26	1297	67.80	28,995.06	1.84	1.69	9,320.23	1.72	4	6
4	HOBBI_1_008_CA_1	HOBBI_DS_1_008	CA_1	CA	HOBBIES	13830	7.23	1378	72.03	6,564.08	0.47	0.38	2,074.50	0.38	5	73
5	FOODS_1_004_WI_1	FOODS_DS_1_004	WI_1	WI	FOODS	13447	7.03	1269	66.34	24,649.08	1.83	1.43	7,933.73	1.46	6	7
6	FOODS_1_004_WI_3	FOODS_DS_1_004	WI_3	WI	FOODS	13422	7.02	1249	65.29	24,299.50	1.81	1.41	7,918.98	1.46	7	8
7	FOODS_1_004_TX_2	FOODS_DS_1_004	TX_2	TX	FOODS	13020	6.81	1217	63.62	23,366.44	1.79	1.36	7,681.80	1.42	8	10
8	FOODS_1_004_TX_3	FOODS_DS_1_004	TX_3	TX	FOODS	12464	6.52	1309	68.43	22,977.74	1.84	1.34	7,353.76	1.36	9	12
9	FOODS_1_004_TX_1	FOODS_DS_1_004	TX_1	TX	FOODS	10798	5.64	1323	69.16	19,763.40	1.83	1.15	6,370.82	1.17	10	16

✅ SKU-store performance validation passed.

Highest-demand series : FOODS_1_004_WI_2 (18,709 units)

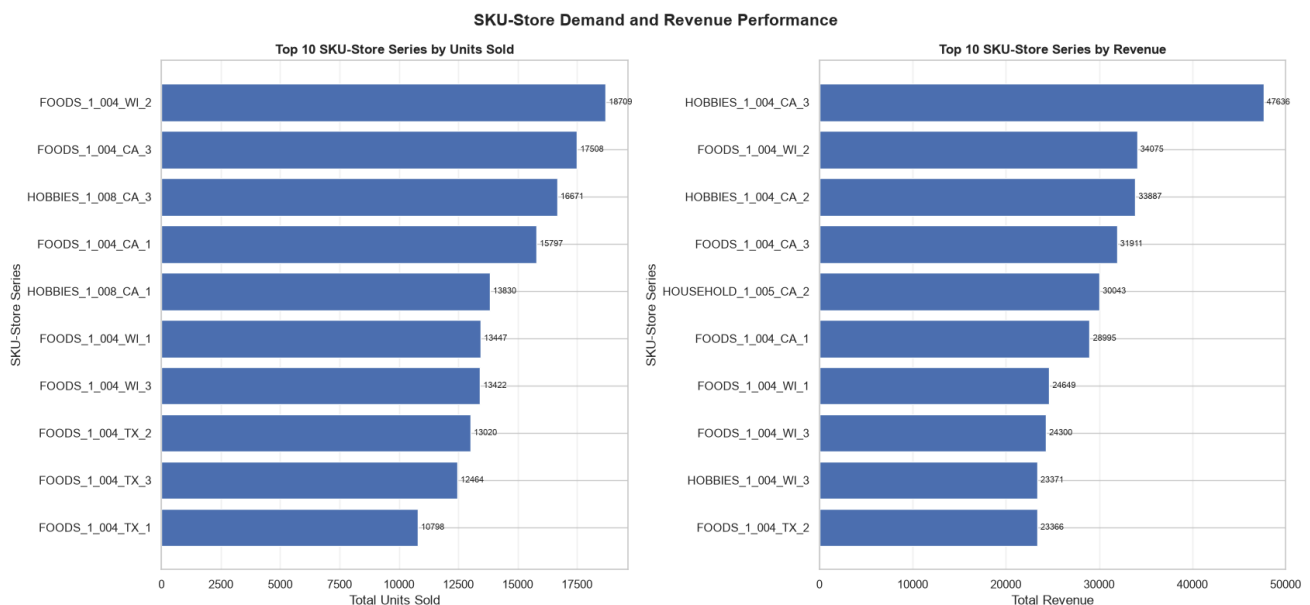
Lowest-demand series : HOBBIES_1_006_CA_4 (25 units)

Highest-revenue series : HOBBIES_1_004_CA_3 (47,635.78)

Lowest-revenue series : HOBBIES_1_006_CA_4 (24.32)

Highest selling frequency: HOBBIES_1_004_CA_3 (86.30%)

Lowest selling frequency : HOBBIES_1_006_CA_4 (1.05%)



✅ SKU-store performance figure created successfully.

Demand leader : FOODS_1_004_WI_2

Revenue leader : HOBBIES_1_004_CA_3

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_6_top_sku_store_performance.png

SKU-Store Performance Summary

- **Demand leader:** FOODS_1_004_WI_2 recorded the highest demand with **18,709 units sold**.
- **Revenue leader:** HOBBIES_1_004_CA_3 generated the highest revenue at **47,635.78**, despite not ranking among the highest-volume series.
- **Strong product consistency:** FOODS_1_004 appears across several stores in the top-demand ranking, confirming strong demand across multiple locations.
- **Location effect:** The performance of the same product varies noticeably across stores, supporting the use of the **SKU-store level** for forecasting and inventory decisions.
- **Highest selling frequency:** HOBBIES_1_004_CA_3 recorded demand on **86.30% of observed days**, supporting its strong revenue performance.
- **Weakest series:** HOBBIES_1_006_CA_4 recorded only **25 units, 24.32 in revenue**, and a selling frequency of **1.05%**.
- **Key insight:** High demand does not always produce the highest revenue because selling price and store-level product mix also influence financial contribution.
- **Business implication:** Prioritise availability for leading SKU-store combinations, use store-specific replenishment rules, and investigate extremely weak series for reduced stocking or

possible removal.

Estimated margin findings remain assumption-based because actual product-cost data was unavailable.

✔ Section 8.6: SKU-Store Performance is complete.

8.7 Contribution and Ranking Analysis

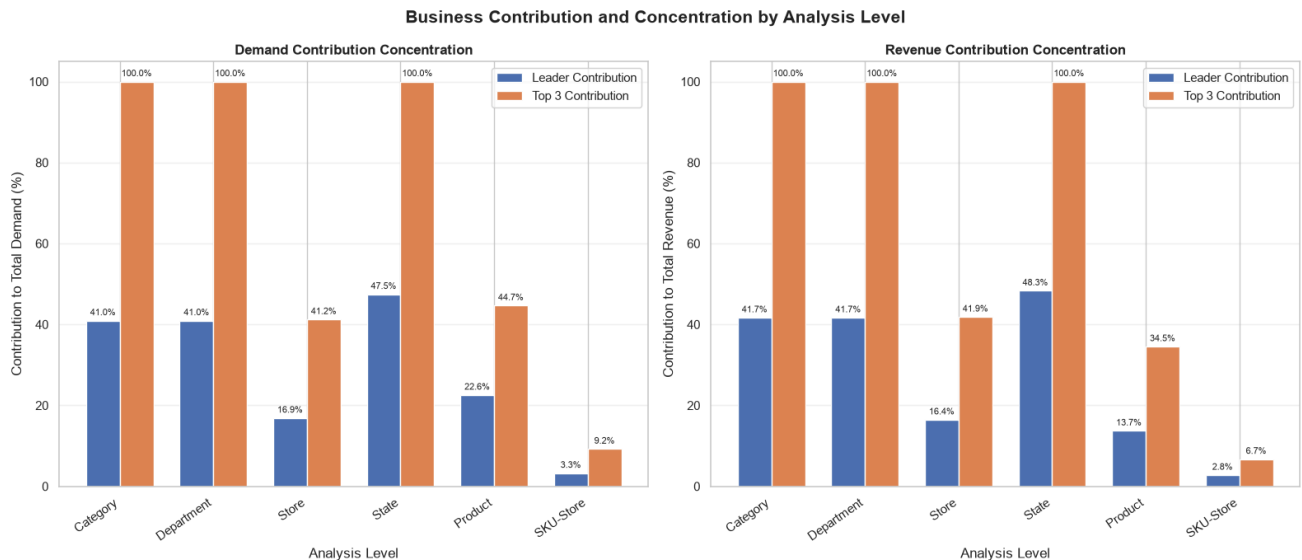
Categories, departments, stores, states, products, and SKU-store series will be ranked using relevant demand and revenue measures.

Contribution percentages will be calculated to show how much each segment contributes to the overall business.

Business contribution summary prepared successfully.
Analysis levels covered : 6
Total entities ranked : 349

	Analysis Level	Entity Count	Demand Leader	Demand Leader Contribution (%)	Revenue Leader	Revenue Leader Contribution (%)	Same Demand and Revenue Leader	Top 3 Demand Contribution (%)	Top 3 Revenue Contribution (%)
0	Category	3	FOODS	40.95	HOUSEHOLD	41.68	False	100.00	100.00
1	Department	3	FOODS_1	40.95	HOUSEHOLD_1	41.68	False	100.00	100.00
2	Store	10	CA_3	16.88	CA_3	16.45	True	41.23	41.86
3	State	3	CA	47.46	CA	48.32	True	100.00	100.00
4	Product	30	FOODS_1_004	22.56	FOODS_1_004	13.72	True	44.66	34.54
5	SKU-Store	300	FOODS_1_004_WI_2	3.26	HOBBIES_1_004_CA_3	2.77	False	9.22	6.72

✔ Business contribution summary validation passed.
Analysis levels validated : 6
Total entities validated : 349
Levels with same leader : 3 of 6
Highest top-3 demand concentration : Category (100.00%)
Highest top-3 revenue concentration: Category (100.00%)



✓ Contribution concentration figure created successfully.

Levels with same leader : 3 of 6

Three-entity levels : Category, Department, State

Figure path : C:\Users\hp\Project FORESIGHT\reports\figures\08_7_contribution_concentration.png

Contribution and Ranking Summary

- **Leader alignment:** Demand and revenue leaders were the same at **3 of 6 levels:** Store, State, and Product.
- **Category and department:** FOODS led demand, while HOUSEHOLD led revenue, confirming the difference between volume and financial value.
- **Store concentration:** The top three stores contributed **41.23% of demand** and **41.86% of revenue**.
- **Product concentration:** The top three products generated **44.66% of demand** but only **34.54% of revenue**.
- **SKU-store distribution:** The top three SKU-store series contributed only **9.22% of demand** and **6.72% of revenue**, showing that performance is widely distributed across locations.
- **Interpretation note:** The 100% top-three values for Category, Department, and State reflect their three-entity structure, not unusually high concentration.
- **Business implication:** Strategic decisions should combine rankings with contribution percentages and should not assume that the demand leader is also the strongest financial contributor.

✓ Section 8.7: Contribution and Ranking Analysis is complete.

8.8 Business Performance Summary

The most important findings will be documented using:

- performance observation;
- supporting evidence;
- business interpretation;
- recommended area for further investigation.

Interpretation rule: High sales volume does not always imply high revenue or high margin. Demand, price, and estimated financial contribution must be considered together.

Business performance summary prepared successfully.
Findings documented : 6
Summary fields : 5

✓ Business performance summary validation passed.

Validated findings : 6

Validated fields : 5

✓ Section 8.8 completed: Business performance findings consolidated.

Business Performance Summary

- **Volume versus value:** FOODS led demand, while HOUSEHOLD led revenue, showing that **high unit sales do not always create the highest financial contribution**.
- **Store leader:** CA_3 ranked first in both demand and revenue, while **CA_4 recorded the weakest store performance**.
- **State leader:** California contributed **47.46% of demand** and also recorded the highest demand per store.
- **Product concentration:** FOODS_1_004 led both product demand and revenue, contributing **22.56% of total units sold**.
- **Location effect:** SKU-store demand and revenue leaders were different, confirming that **product performance varies by store**.
- **Weak-series risk:** HOBBIES_1_006_CA_4 sold only **25 units** and was active on **1.05% of observed days**.
- **Business priority:** Protect high-performing products and locations, apply SKU-store-specific planning, and investigate weak series for reduced stocking or removal.

Revenue and demand results are based on historical observations. Estimated-margin findings remain assumption-based because actual unit-cost data was unavailable.

✓ Section 8.8: Business Performance Summary is complete.

✓ Section 8: Business Performance Analysis is complete.

9. Top Movers, Slow Movers, and Dead-Stock Candidates

This section converts the performance findings from Section 8 into practical product-movement and inventory-risk classifications.

The analysis focuses on:

- historical and recent demand;
- sales frequency and recency;
- fast- and slow-moving products;
- potential dead-stock risk at the SKU-store level;
- simulated inventory exposure.

Product demand and revenue rankings are **reused from Section 8.5** rather than calculated and visualised again.

Business Question

Which products and SKU-store series require stronger availability, reduced replenishment, or further inventory review?

Objectives

- classify products using transparent historical and recent-demand rules;
- measure product sales concentration;
- identify fast-, regular-, and slow-moving products;
- evaluate potential dead-stock risk at the SKU-store level;
- estimate the simulated inventory exposure of weak-moving series.

Analysis Structure

- **9.1:** Define the 90-day analysis window and classification rules.
- **9.2:** Measure cumulative demand and revenue concentration.
- **9.3:** Summarise product movement classifications.
- **9.4:** Evaluate dead-stock candidates and simulated inventory exposure.
- **9.5:** Consolidate the main findings and business implications.

Interpretation rule: Movement classifications are analytical screening labels, not final assortment decisions. Inventory quantities and values are simulated and must not be presented as observed financial losses.

9.1 Define the Analysis Window and Classification Rules

Product movement is evaluated using both **full-history performance** and the most recent **90 calendar days**.

The analysis uses:

- historical and recent demand;
- active and zero-demand days;
- selling frequency;
- days since the most recent sale;
- latest simulated inventory quantity and value.

Classification thresholds are derived from the observed product-demand distribution. A stricter rule is used for dead-stock candidates: **no sales during the full 90-day window, at least 90 days since the last sale, and positive simulated inventory**.

These classifications are screening indicators for further investigation, not final replenishment or assortment decisions.

Movement analysis window prepared successfully.
Full-history period : 2011-01-29 to 2016-04-24
Recent period : 2016-01-26 to 2016-04-24
Recent days covered : 90
Products represented: 30
SKU-store series : 300

	Movement Metric	Minimum	25th Percentile	Median	75th Percentile	Maximum
0	total_units_sold	1,435.00	6,356.50	11,300.00	17,463.75	129,481.00
1	average_daily_demand	0.75	3.32	5.91	9.13	67.68
2	recent_units_sold	86.00	412.50	551.50	952.00	3,585.00
3	recent_average_daily_demand	0.96	4.58	6.13	10.58	39.83
4	recent_selling_day_frequency_pct	17.78	97.78	98.89	100.00	100.00
5	recent_zero_demand_frequency_pct	0.00	0.00	1.11	2.22	82.22
6	days_since_last_sale	0.00	0.00	0.00	0.00	53.00
7	latest_closing_stock	18.00	63.25	109.00	186.25	1,864.00
8	latest_inventory_value	41.64	181.83	308.66	501.60	2,553.68

✔ Product movement indicators prepared successfully.
Products analysed : 30
Recent window : 90 days
Latest inventory value: 14,170.55

	Movement Classification	Product Count
0	Regular-Moving	19
1	Fast-Moving	7
2	Slow-Moving	4

✓ Movement classification rules applied successfully.

Fast history threshold : 17,463.75 units

Fast recent threshold : 10.58 units/day

Fast frequency threshold : 98.89%

Slow history threshold : 6,356.50 units

Slow recent threshold : 4.58 units/day

Slow frequency threshold : 98.89%

Dead-stock rule : No sales during the full 90-day window with positive simulated stock

Analysis Window and Classification Rules Summary

- **Recent window:** Final **90 calendar days**, from 26 January 2016 to 24 April 2016.
- **Fast-moving rule:** Strong historical demand, strong recent demand, and frequent recent sales.
- **Slow-moving rule:** Low historical demand, low recent demand, and below-median recent selling frequency.
- **Dead-stock rule:** No demand during the complete 90-day window, at least 90 days since the last sale, and positive simulated inventory.
- **Classification result:** **7 Fast-Moving**, **19 Regular-Moving**, and **4 Slow-Moving products**.
- **Dead-stock result:** No product met the strict product-level dead-stock rule.
- **Modelling note:** Product-level selling frequency is generally high because demand is combined across ten stores. Dead-stock risk will therefore be assessed more precisely at the **SKU-store level** in Section 9.4.

✓ Section 9.1: Analysis Window and Movement Rules is complete.

9.2 Sales Concentration Analysis

This analysis measures how much of total demand and revenue is generated by the leading products.

Existing product rankings from Section 8.5 are reused to avoid repeating the same tables and charts.

Products are arranged from highest to lowest contribution, and cumulative percentages are calculated to identify:

- the number of products responsible for approximately **80% of demand**;
- the number responsible for approximately **80% of revenue**;
- whether business performance is concentrated in a small part of the product portfolio.

■ The 80% level is used as a descriptive Pareto reference, not as a fixed business rule.

✓ Product concentration curves prepared successfully.

Products analysed : 30

Metrics prepared : 2

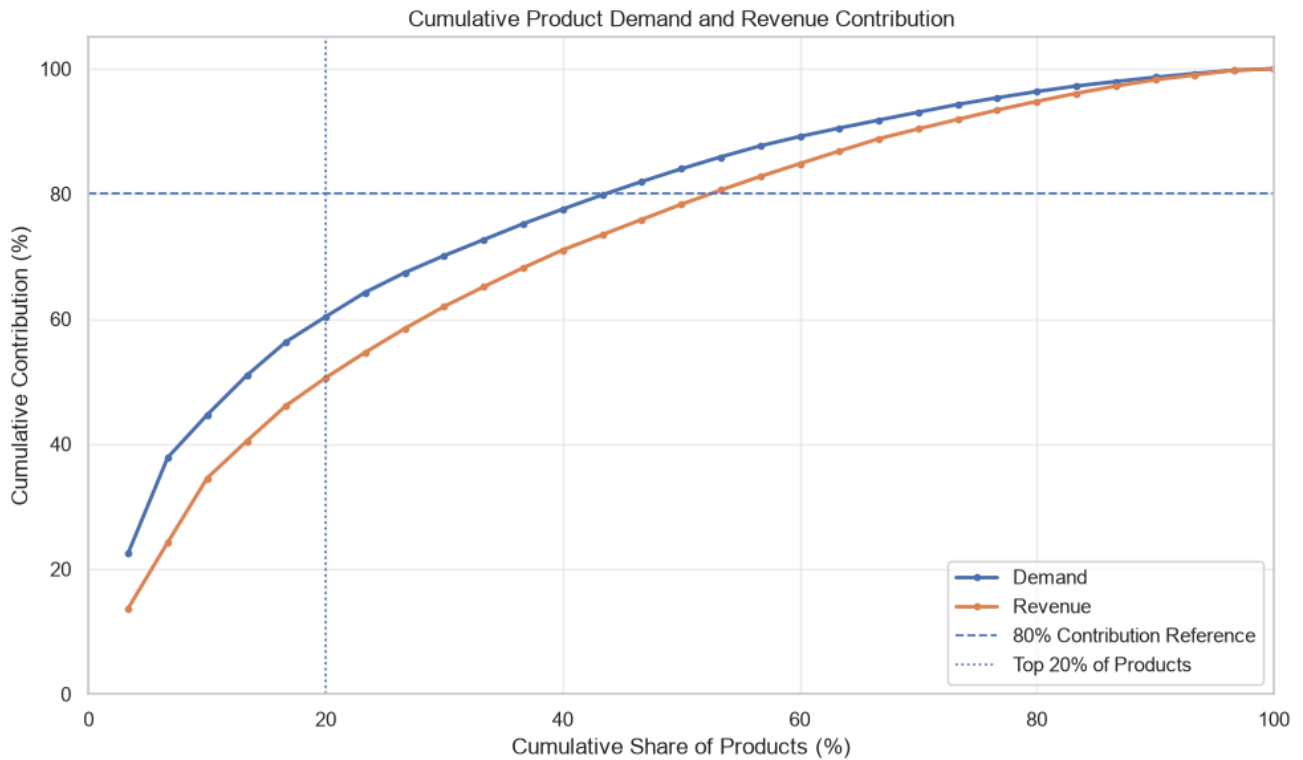
Curve records : 60

	Performance Metric	Products to Reach 80%	Portfolio Share to Reach 80% (%)	Contribution at Cutoff (%)	Top 20% Product Count	Top 20% Contribution (%)
0	Demand	14	46.67	81.95	6	60.35
1	Revenue	16	53.33	80.59	6	50.58

✅ Product concentration levels summarised successfully.

Pareto reference : 80%

Top portfolio group : 6 of 30 products



✅ Product concentration figure created successfully.

Figure saved to: C:\Users\hp\Project FORESIGHT\reports\figures\09_2_product_sales_concentration.png

Sales Concentration Findings

- **Demand concentration:** The top **6 products**, representing 20% of the portfolio, generated **60.35% of total demand**.
- **Revenue concentration:** The same top-20% product group generated **50.58% of total revenue**.
- **80% demand threshold:** **14 products**, or 46.67% of the portfolio, were required to reach **81.95% of demand**.
- **80% revenue threshold:** **16 products**, or 53.33% of the portfolio, were required to reach **80.59% of revenue**.
- **Interpretation:** Demand is more concentrated than revenue, but the portfolio does not follow a strict 80/20 pattern.
- **Business implication:** Leading products require stronger availability controls, while the broader portfolio still contributes materially and should not be ignored.

✔ Section 9.2: Sales Concentration Analysis is complete.

9.3 Product Movement Classification

Products are grouped into **Fast-Moving**, **Regular-Moving**, and **Slow-Moving** classes using the rules established in Section 9.1.

The comparison will focus on:

- historical and recent demand;
- recent selling frequency;
- movement-class distribution;
- category and department composition;
- inventory-planning implications.

Only one consolidated classification table and one supporting figure will be used.

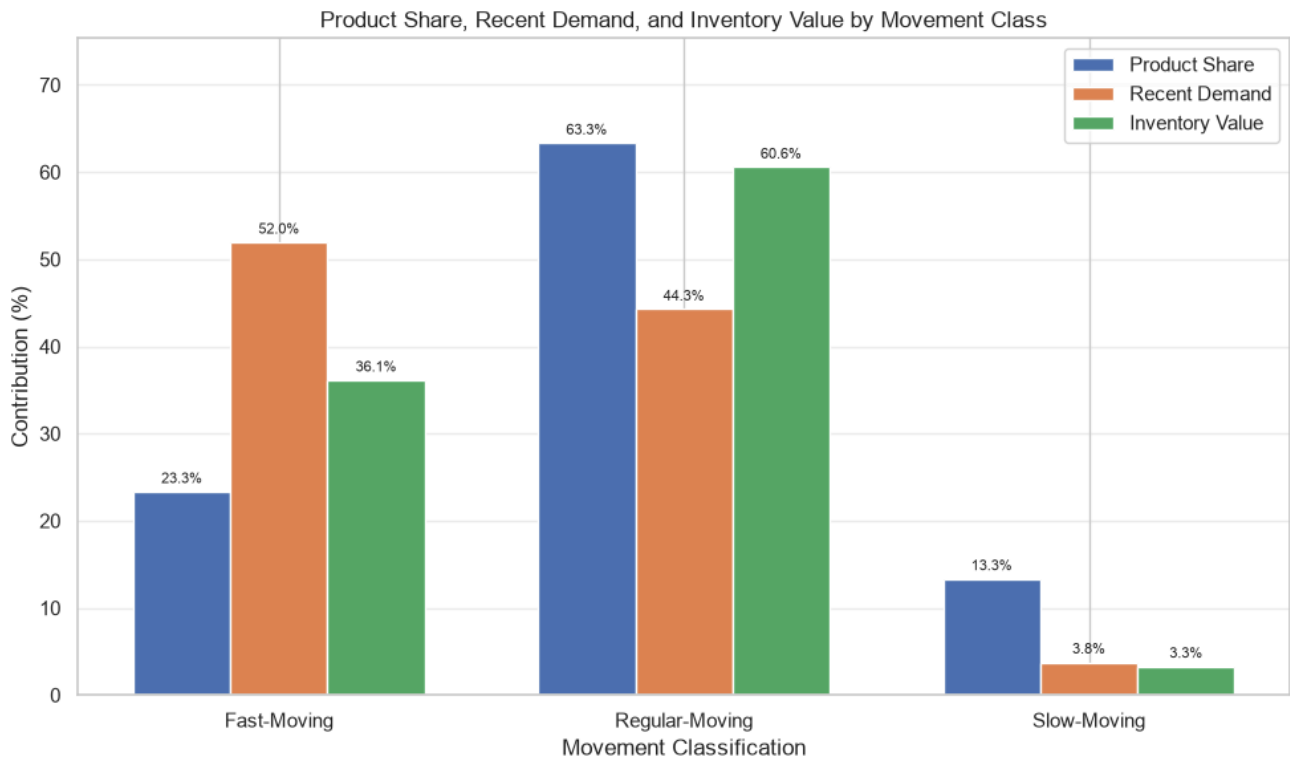
Movement classes are relative screening labels based on the current analytical scope and should be reviewed periodically as demand changes.

	Movement Class	Products	Product Share (%)	Historical Demand (%)	Recent Demand (%)	Revenue (%)	Inventory Value (%)	Avg Recent Daily Demand	Avg Recent Selling Frequency (%)	Avg Days Since Last Sale
0	Fast-Moving	7	23.33	44.82	51.95	37.73	36.13	20.51	100.00	0.00
1	Regular-Moving	19	63.33	52.83	44.27	58.71	60.58	6.44	89.94	2.79
2	Slow-Moving	4	13.33	2.34	3.77	3.56	3.29	2.61	90.83	0.00

✔ Product movement classes summarised successfully.

Movement classes represented: 3

Products classified : 30



✅ Product movement classification figure created successfully.

Figure saved to: C:\Users\hp\Project FORESIGHT\reports\figures\09_3_product_movement_classification.png

Product Movement Classification Findings

- **Fast-Moving products: 7 products** represented **23.33% of the portfolio** but generated **51.95% of recent demand**.
- **Regular-Moving products: 19 products** represented **63.33% of the portfolio** and held **60.58% of simulated inventory value**.
- **Slow-Moving products: 4 products** represented **13.33% of the portfolio** but contributed only **3.77% of recent demand** and **3.29% of simulated inventory value**.
- **Demand concentration:** Fast-moving products generated more than half of recent demand despite representing less than one-quarter of the product portfolio.
- **Inventory implication:** Fast-moving products require stronger availability protection, while slow-moving products should be reviewed for reduced replenishment or targeted promotion.
- **Modelling note:** Product-level recency remains high because sales are aggregated across stores. Location-specific inventory risk will therefore be assessed at the **SKU-store level** in Section 9.4.

✅ Section 9.3: Product Movement Classification is complete.

9.4 Dead-Stock Candidates and Inventory Exposure

Dead-stock risk is evaluated at the **SKU-store level**, where weak product movement can be hidden by stronger sales in other locations.

Candidate identification will consider:

- recent demand;
- selling frequency;
- days since the most recent sale;
- positive simulated closing stock;
- estimated simulated inventory value.

The analysis will also summarise inventory exposure by store, state, and product where candidates are identified.

Dead-stock labels are preliminary screening indicators. Inventory quantities and values are simulated and must not be presented as confirmed financial loss.

	Screening Metric	Minimum	25th Percentile	Median	75th Percentile	Maximum
0	recent_units_sold	2.00	26.75	56.50	101.25	953.00
1	recent_active_selling_days	1.00	19.00	34.00	50.25	87.00
2	recent_selling_frequency_pct	1.11	21.11	37.78	55.83	96.67
3	recent_zero_demand_frequency_pct	3.33	44.17	62.22	78.89	98.89
4	days_since_last_sale	0.00	0.00	1.00	3.00	86.00
5	closing_stock	0.00	4.00	9.00	19.25	335.00
6	estimated_inventory_value	0.00	11.10	25.80	53.76	471.25

✓ SKU-store movement indicators prepared successfully.

SKU-store series analysed : 300

Recent analysis window : 90 days

Series with zero recent demand: 0

Series inactive for 90+ days : 0

	Inventory Risk Status	SKU-Store Series	Series Share (%)	Recent Units	Simulated Inventory Units	Estimated Inventory Value	Inventory Value Share (%)	Avg Recent Units	Avg Selling Frequency (%)	Avg Days Since Last Sale
0	Weak-Movement Watchlist	63	21.00	856	572	1,417.05	10.00	13.59	11.76	17.87
1	Active / No Immediate Risk	237	79.00	24016	5912	12,753.50	90.00	101.33	46.98	3.92

✓ SKU-store inventory-risk screening completed.

Weak recent-units threshold : 26.75

Weak frequency threshold : 21.11%

Strict dead-stock candidates: 0

Weak-movement watchlist : 63

	SKU-Store Series	Product	Store	State	Recent Units	Recent Selling Frequency (%)	Days Since Last Sale	Simulated Closing Stock	Estimated Inventory Value	Watchlist Value Share (%)
0	FOODS_1_004_TX_2	FOODS_1_004	TX_2	TX	25	5.56	85	157	215.09	15.18
1	FOODS_1_002_WI_3	FOODS_1_002	WI_3	WI	13	13.33	8	9	59.76	4.22
2	HOUSEHOLD_1_002_CA_1	HOUSEHOLD_1_002	CA_1	CA	3	3.33	0	11	53.68	3.79
3	FOODS_1_002_TX_1	FOODS_1_002	TX_1	TX	8	8.89	8	8	53.12	3.75
4	HOUSEHOLD_1_009_WI_1	HOUSEHOLD_1_009	WI_1	WI	18	18.89	8	4	50.32	3.55
5	HOUSEHOLD_1_001_TX_2	HOUSEHOLD_1_001	TX_2	TX	11	12.22	2	10	48.80	3.44
6	FOODS_1_004_CA_4	FOODS_1_004	CA_4	CA	11	4.44	86	34	46.58	3.29
7	HOUSEHOLD_1_002_TX_1	HOUSEHOLD_1_002	TX_1	TX	23	21.11	2	8	39.04	2.76
8	HOUSEHOLD_1_001_TX_1	HOUSEHOLD_1_001	TX_1	TX	22	21.11	2	8	39.04	2.76
9	FOODS_1_009_CA_3	FOODS_1_009	CA_3	CA	16	11.11	35	19	35.72	2.52

✅ Weak-movement inventory exposure ranked successfully.

Watchlist SKU-store series : 63

Simulated inventory units : 572

Estimated inventory value : 1,417.05

Dead-Stock and Inventory Exposure Findings

- **Strict dead-stock candidates:** No SKU-store series met the strict rule because every series recorded some demand during the recent 90-day period.
- **Weak-movement watchlist: 63 SKU-store series**, representing **21% of all series**, showed both low recent demand and low selling frequency while holding positive simulated inventory.
- **Simulated exposure:** Watchlist series held **572 inventory units** with an estimated value of **1,417.05**, representing **10% of total simulated inventory value**.
- **Largest individual exposure:** FOODS_1_004_TX_2 held **157 simulated units** valued at **215.09**, accounting for **15.18% of watchlist value**.
- **Recency concern:** FOODS_1_004_TX_2 and FOODS_1_004_CA_4 had recorded no sale for **85 and 86 days**, respectively, placing them close to the strict 90-day inactivity threshold.
- **Business implication:** Watchlist series should be reviewed for reduced replenishment, stock transfer, targeted promotion, or assortment adjustment rather than being automatically classified as dead stock.

Inventory quantities and values are simulated. The estimated exposure represents an analytical screening measure, not confirmed financial loss.

✓ Section 9.4: Dead-Stock Candidates and Inventory Exposure is complete.

9.5 Movement Classification Summary

This subsection consolidates the main findings from the movement and inventory-risk analysis.

The summary will document:

- product sales concentration;
- fast-, regular-, and slow-moving product counts;
- leading and weak movement patterns;
- dead-stock screening results;
- simulated inventory exposure;
- recommended areas for further investigation.

The final output will use one concise findings table and a short bullet-point interpretation.

✓ Movement classification findings consolidated.

Validated findings : 6

✓ Section 9.5 completed: Movement classification summary prepared.

Movement Classification Summary

- **Sales concentration:** The top 20% of products generated **60.35% of demand** and **50.58% of revenue**, indicating stronger concentration in unit demand than financial value.
- **Fast-moving products: 7 products** generated **51.95% of recent demand** and require stronger forecasting and availability protection.
- **Slow-moving products: 4 products** contributed only **3.77% of recent demand** and should be reviewed for replenishment, promotion, or assortment action.
- **Dead-stock screening:** No SKU-store series met the strict 90-day dead-stock rule.
- **Weak-movement watchlist: 63 SKU-store series** held **572 simulated inventory units** with an estimated value of **1,417.05**.
- **Highest-priority series:** FOODS_1_004_TX_2 recorded the largest watchlist exposure, with **157 simulated units valued at 215.09**.
- **Business priority:** Protect fast-moving availability while reviewing weak location-specific inventory before applying transfers, markdowns, or replenishment reductions.

Movement classifications are analytical screening labels. Inventory quantities and values are simulated and do not represent confirmed financial loss.

- ✓ Section 9.5: Movement Classification Summary is complete.
- ✓ Section 9: Top Movers, Slow Movers, and Dead-Stock Candidates is complete.

10. Demand Drivers

This section examines whether available business factors are associated with changes in demand and revenue.

The analysis focuses on:

- selling-price behaviour;
- promotional periods;
- calendar events;
- state-specific SNAP activity;
- variables that may be useful for later forecasting feature engineering.

Only focused and business-relevant comparisons are included. Observed differences and correlations are interpreted as **associations**, not proof that a factor caused demand to change.

Business Question

Which available factors appear to be associated with changes in customer demand, and which should be considered in later forecasting models?

Objectives

- examine price variation and its relationship with demand;
- compare promotional and non-promotional performance;
- evaluate event-day and regular-day demand;
- compare SNAP-active and SNAP-inactive periods;
- identify useful forecasting features and their interpretation limits.

Analysis Structure

- **10.1:** Define the demand-driver fields and analytical rules.
- **10.2:** Examine price behaviour and its relationship with demand.
- **10.3:** Compare promotion, event, and SNAP demand patterns.
- **10.4:** Screen the available variables for forecasting usefulness.
- **10.5:** Consolidate the principal demand-driver findings.

Interpretation rule: Group differences and correlations describe historical associations. They must not be presented as evidence of causation.

10.1 Analysis Scope and Driver Definitions

Demand-driver analysis is performed using the integrated daily SKU-store dataset.

The principal explanatory variables are:

- selling price;
- promotion indicator;
- event indicator;
- SNAP indicator applicable to each state.

The main response measures are:

- units sold;
- revenue;
- selling-day frequency.

Comparisons will use consistent definitions and appropriate aggregation levels to reduce distortion from product and store differences.

Simulated inventory measures are excluded from the core driver analysis because they are evaluated separately in Section 11 and should not be interpreted as independent observed demand drivers.

	Driver	Field	Active Records	Active Share (%)
0	Promotion	promo_flag	2102	0.37
1	Calendar Event	event_flag	46200	8.05
2	State-Specific SNAP	snap_flag	283500	49.40

✓ Demand-driver analytical scope prepared.

Daily SKU-store records : 573,900

Products represented : 30

SKU-store series : 300

Date range : 2011-01-29 to 2016-04-24

Demand-Driver Scope Summary

- **Analysis base:** The driver analysis covers **573,900 daily SKU-store records**, representing **30 products** and **300 SKU-store series**.
- **Promotion coverage:** Promotional periods account for only **0.37% of records**, indicating limited promotional exposure in the available dataset.
- **Event coverage:** Calendar events occur in **8.05% of records**, providing sufficient observations for a broad event-versus-regular-day comparison.
- **SNAP coverage:** State-specific SNAP activity applies to **49.40% of records**, supporting balanced SNAP-active and SNAP-inactive comparisons.
- **Interpretation caution:** Promotion results may be less stable than event and SNAP findings because promotional observations are comparatively rare.

✓ Section 10.1: Analysis Scope and Driver Definitions is complete.

10.2 Price Behaviour and Demand Relationship

This subsection combines price distribution, price changes, and price-demand analysis into one focused workflow.

The analysis will examine:

- typical selling-price levels and variation;
- SKU-store price stability;
- frequency of recorded price changes;
- demand across comparable price conditions;
- the relationship between price and units sold.

Price-demand relationships will be interpreted carefully because an overall correlation may reflect differences between products rather than customer response to price changes.

Within-product or SKU-store comparisons are more informative than comparing unrelated high-price and low-price products.

	Price Measure	Value
0	Minimum Valid Price	0.10
1	Median Valid Price	3.17
2	Average Valid Price	4.38
3	Sales-Weighted Average Price	3.00
4	Maximum Valid Price	17.97
5	Median SKU-Store Price Change Frequency (%)	0.10
6	Valid Price Coverage (%)	85.74

✔ Valid-price behaviour measures prepared successfully.

Zero-price records excluded : 81,823

Valid price records : 492,077

SKU-store series analysed : 300

Series with price changes : 255

Unique valid prices : 133

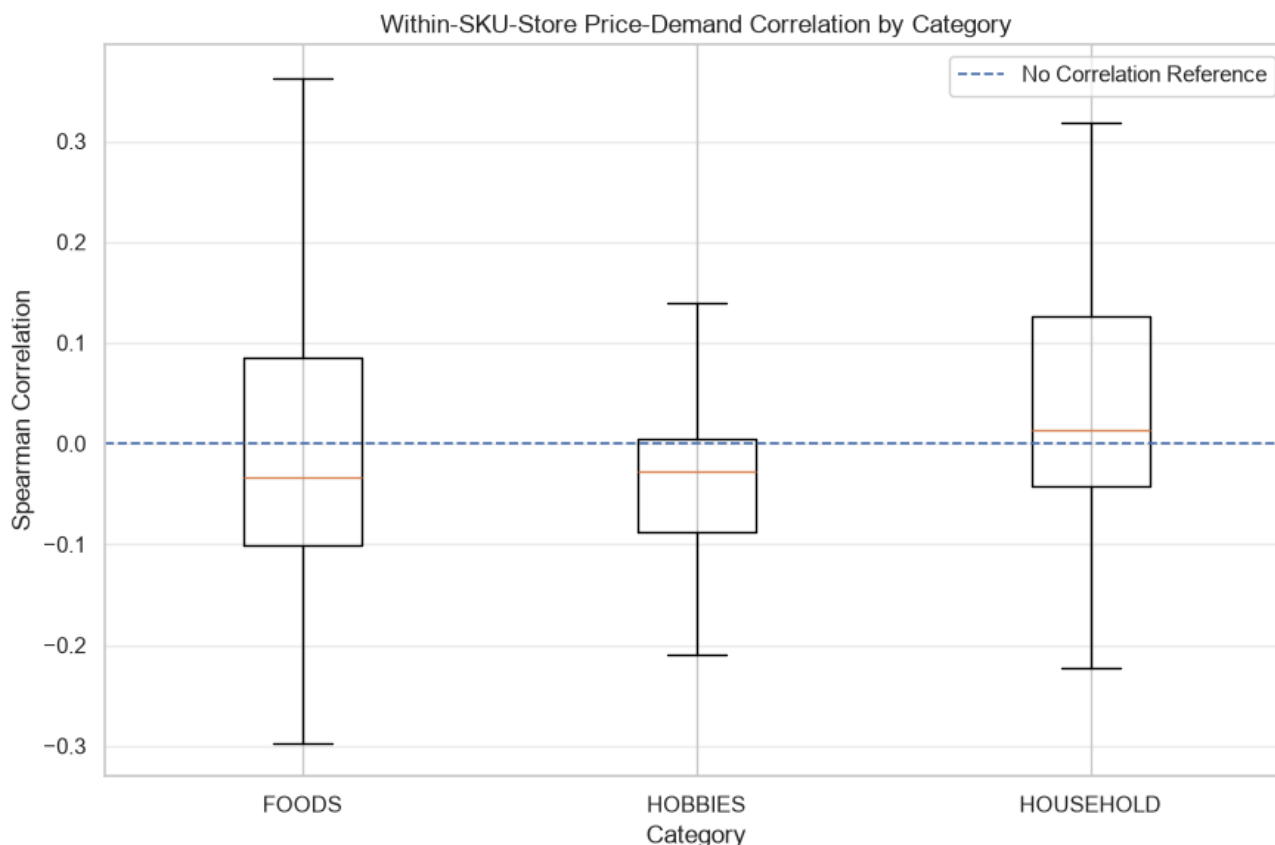
	Category	Products	SKU-Store Series	Minimum Price	Median Price	Average Price	Sales-Weighted Price	Maximum Price	Median Price Variation (%)	Median Price-Change Frequency (%)
0	FOODS	10	100	0.10	2.88	3.60	2.49	9.48	5.33	0.18
1	HOBBIES	10	100	0.42	2.97	3.39	2.13	9.58	1.87	0.07
2	HOUSEHOLD	10	100	1.00	5.97	6.25	5.00	17.97	2.58	0.06

	Relationship Measure	Value
0	SKU-Store Series with Valid Correlation	255.00
1	Median Within-Series Correlation	-0.02
2	Series with Negative Correlation (%)	58.43
3	Series with Positive Correlation (%)	41.57

✔ Category price behaviour and price-demand relationships prepared.

Valid within-series correlations: 255

	Category	Valid SKU-Store Series	Median Correlation	25th Percentile	75th Percentile	Negative Relationship Share (%)
0	FOODS	100	-0.03	-0.10	0.09	57.00
1	HOBBIES	83	-0.03	-0.09	0.00	71.08
2	HOUSEHOLD	72	0.01	-0.04	0.13	45.83



✅ Price-demand relationship figure created successfully.

Figure saved to: C:\Users\hp\Project FORESIGHT\reports\figures\10_2_price_demand_relations
hip_by_category.png

Price Behaviour and Demand Findings

- **Valid price coverage:** Positive selling prices were available for **85.74% of daily SKU-store records**. Zero-price records were excluded only from price analysis because they contained no sales or revenue.
- **Price stability:** Although **255 of 300 SKU-store series** recorded at least one price change, the median price-change frequency was only **0.10%**, indicating generally stable prices.
- **Price levels:** HOUSEHOLD had the highest typical prices, with a median price of **5.97** and a sales-weighted price of **5.00**.
- **Sales mix:** The overall sales-weighted price of **3.00** was below the simple average valid price of **4.38**, showing that demand was concentrated more heavily among lower-priced products.
- **Overall relationship:** The median within-SKU-store price-demand correlation was **-0.02**, indicating almost no consistent relationship across the portfolio.
- **Category patterns:** FOODS and HOBBIES had slightly negative median correlations of **-0.03**, while HOUSEHOLD had a near-zero positive median correlation of **0.01**.
- **Modelling implication:** Price should be retained as a forecasting feature, but its effect should be evaluated at the product or SKU-store level rather than represented by one portfolio-wide relationship.

These correlations describe historical association. They do not estimate price elasticity or prove that price changes caused demand changes.

✓ Section 10.2: Price Behaviour and Demand Relationship is complete.

10.3 Promotion, Event, and SNAP Demand Comparisons

Promotional periods, calendar events, and state-specific SNAP activity are evaluated in one consolidated driver comparison.

For each driver, the analysis will compare:

- average units sold;
- average revenue;
- selling-day frequency;
- percentage difference between active and inactive periods.

The results will identify whether these periods coincide with meaningful demand differences and whether the variables may be useful for forecasting.

Higher demand during an active period does not prove that the promotion, event, or SNAP programme caused the increase.

	Driver	Active Records	Active Share (%)	Inactive Avg Units	Active Avg Units	Demand Difference (%)	Inactive Avg Revenue	Active Avg Revenue	Revenue Difference (%)	Inactive Selling Frequency (%)	Active Selling Frequency (%)	Selling Frequency Difference (pp)
0	Promotion	2102	0.37	1.00	1.17	17.34	3.00	2.46	-17.95	33.63	33.02	-0.62
1	Calendar Event	46200	8.05	1.00	0.96	-4.81	3.01	2.84	-5.54	33.84	31.26	-2.58
2	State-Specific SNAP	283500	49.40	0.98	1.02	4.94	2.95	3.04	3.01	33.37	33.90	0.53

✓ Promotion, event, and SNAP comparisons prepared.

Drivers compared : 3

Response measures: average units, average revenue, and selling frequency

Promotion, Event, and SNAP Findings

- **Promotion:** Promotional records showed **17.34% higher average units sold**, but **17.95% lower average revenue** than non-promotional records.
- **Promotion limitation:** Promotions covered only **0.37% of observations**, so the apparent demand increase should be treated as preliminary rather than a stable portfolio-wide effect.
- **Calendar events:** Event-day average demand was **4.81% lower**, average revenue was **5.54% lower**, and selling frequency declined by **2.58 percentage points**.

- **SNAP activity:** SNAP-active periods recorded **4.94% higher average demand**, **3.01% higher average revenue**, and a **0.53 percentage-point increase** in selling frequency.
- **Forecasting implication:** Event and SNAP indicators have sufficient coverage for later forecasting tests. Promotion may still be useful, but sparse observations require careful validation and possibly product-level treatment.

These results describe historical group differences. They do not prove that promotions, events, or SNAP activity caused the observed demand changes.

✓ Section 10.3: Promotion, Event, and SNAP Demand Comparisons is complete.

10.4 Demand-Driver Feature Screening

The available variables will be screened for possible use in later forecasting models.

Each driver will be reviewed using:

- observed demand difference;
- consistency of the relationship;
- data coverage;
- business interpretation;
- recommended modelling treatment;
- important limitations.

A broad correlation heatmap is not required. Focused driver-level evidence is more interpretable because the explanatory variables include both numerical and binary fields.

The output will classify each variable as:

- suitable for forecasting;
- suitable with caution;
- requiring transformation or interaction terms;
- unsuitable as a direct explanatory feature.

	Feature	Data Coverage	Observed Evidence	Screening Decision	Recommended Treatment	Important Limitation
0	Selling Price	85.74% valid-price records	Median within-SKU-store correlation: -0.02	Suitable with caution	Use at product or SKU-store level; consider pr...	Relationship is weak and inconsistent; zero-pr...
1	Promotion Flag	0.37% active records	17.34% demand difference	Suitable with caution	Retain as a binary feature and validate by pro...	Promotion observations are very sparse and rev...
2	Calendar Event Flag	8.05% active records	-4.81% demand difference	Suitable for forecasting tests	Retain the event flag and later test event-nam...	Different events may have opposite effects, wh...
3	State-Specific SNAP Flag	49.40% active records	4.94% demand difference	Suitable for forecasting tests	Use as a state-aware binary feature and test i...	Observed differences may also reflect state, c...

✅ Demand-driver features screened successfully.

Features reviewed: 4

Forecasting-test candidates: 2

Features requiring caution: 2

Demand-Driver Feature Screening Findings

- **Selling price:** Retain as a forecasting feature with caution. Its portfolio-wide relationship with demand is weak, so it should be modelled at the product or SKU-store level.
- **Promotion flag:** Retain for testing, but treat results cautiously because promotional observations represent only **0.37% of records**.
- **Calendar event flag:** Suitable for forecasting tests because it has sufficient coverage and event periods showed a measurable historical demand difference.
- **State-specific SNAP flag:** Suitable for forecasting tests because it has balanced coverage and SNAP-active periods were associated with moderately higher demand.
- **Feature-engineering priority:** Test event and SNAP indicators directly, while evaluating price and promotion using product-, category-, or store-specific interactions.
- **Modelling caution:** These screening decisions identify potentially useful predictors; they do not confirm future predictive performance.

✅ Section 10.4: Demand-Driver Feature Screening is complete.

10.5 Demand-Driver Summary

This subsection consolidates the principal findings from the price, promotion, event, and SNAP analyses.

The final summary will document:

- the observed relationship;
- supporting evidence;
- business interpretation;
- potential forecasting implication;
- limitation or caution.

Only one concise findings table and a short Markdown interpretation will be included.

✅ Demand-driver findings consolidated successfully.

Validated findings : 5

✅ Section 10.5 completed: Demand-driver summary prepared.

Demand-Driver Summary

- **Selling price:** Price showed a weak and inconsistent relationship with demand, with a median within-SKU-store correlation of **-0.02**.

- **Promotion:** Promotional observations recorded **17.34% higher average demand** but **17.95% lower average revenue**. This result requires caution because promotions covered only **0.37% of records**.
- **Calendar events:** Event periods were associated with **4.81% lower average demand** and a **2.58 percentage-point reduction** in selling frequency.
- **SNAP activity:** SNAP-active periods recorded **4.94% higher average demand** and **3.01% higher average revenue**.
- **Forecasting priority:** Event and state-specific SNAP indicators are suitable for direct forecasting tests. Price and promotion should be retained with product-, store-, or category-level treatment.
- **Validation requirement:** Final feature usefulness must be evaluated through time-based forecasting experiments rather than EDA relationships alone.

All findings describe historical associations. They do not establish causal effects or guarantee future predictive performance.

✓ Section 10.5: Demand-Driver Summary is complete.

✓ Section 10: Demand Drivers is complete.



11. Inventory and Demand Behaviour

This section examines simulated inventory conditions together with historical SKU-store demand behaviour.

The analysis focuses on:

- simulated inventory availability and coverage;
- low-stock, stockout, and excess-stock conditions;
- demand variability and intermittency;
- combined inventory-demand patterns relevant to forecasting and planning.

Weak-movement and dead-stock screening from Section 9 will be reused where relevant rather than recalculated.

Business Question

Which SKU-store series show inventory concern, unstable demand, or intermittent sales behaviour that may affect forecasting and inventory planning?

Objectives

- summarise simulated inventory position and status;
- evaluate inventory coverage using historical demand;

- measure SKU-store demand variability and intermittency;
- classify practical demand-behaviour segments;
- combine inventory and demand signals into preliminary planning groups;
- document forecasting and inventory implications without creating final risk scores.

Analysis Structure

- **11.1:** Define the inventory and demand-behaviour scope.
- **11.2:** Summarise simulated inventory status and coverage.
- **11.3:** Measure demand variability and intermittency.
- **11.4:** Combine inventory and demand behaviour into planning segments.
- **11.5:** Consolidate forecasting and inventory implications.

Scope boundary: Inventory records are simulated. This section provides descriptive screening only; final forecasts, risk scores, reorder quantities, and recommended actions will be developed in later notebooks.

11.1 Analysis Scope and Inventory Measures

The analysis uses the integrated daily SKU-store dataset and the latest simulated inventory snapshot.

The available inventory measures include:

- closing stock;
- inventory on order;
- reorder point and reorder quantity;
- lead time;
- target stock;
- inventory status;
- estimated inventory value.

Inventory coverage will be derived using the latest simulated closing stock and recent SKU-store demand rather than relying on a pre-existing coverage field.

Historical demand behaviour will later be evaluated using average demand, variability, selling frequency, zero-demand frequency, recent demand, and sales recency.

Latest-snapshot inventory measures are kept separate from full-history demand measures to avoid mixing different analytical time levels.

	Validation Check	Result
0	Latest Inventory Date	2016-04-24
1	Latest Snapshot Records	300
2	Unique SKU-Store Series	300
3	Duplicate SKU-Store Series	0
4	Missing Critical Values	0
5	Inventory Status Levels	4

✓ Latest simulated inventory scope prepared.
Inventory statuses: Healthy, Low Stock, Overstock, Stockout

Inventory Analysis Scope Summary

- **Snapshot date:** Inventory analysis uses the latest simulated snapshot dated **24 April 2016**.
- **Analytical grain:** The snapshot contains exactly **300 unique SKU-store series**, with one record per series.
- **Data quality:** No duplicate series or missing values were found in the critical inventory fields.
- **Available statuses:** The simulated inventory records contain four conditions: **Healthy, Low Stock, Overstock, and Stockout**.
- **Scope control:** This subsection validates the inventory-analysis base only. Inventory quantities, values, and status distributions will be examined in Section 11.2.

All inventory measures are simulated and represent analytical scenarios rather than observed retailer inventory.

✓ Section 11.1: Analysis Scope and Inventory Measures is complete.

11.2 Simulated Inventory Status and Coverage

This subsection combines the inventory overview, coverage analysis, low-stock observations, stockouts, and excess-stock observations into one workflow.

The analysis will summarise:

- latest simulated inventory quantity and value;
- low-stock, stockout, normal, and excess-stock statuses;
- inventory coverage distribution;
- concentration by category and store;
- demand associated with each inventory condition.

Only one consolidated status table and one supporting figure will be used.

Inventory coverage and status are descriptive simulation outputs and must not be presented as observed retailer inventory performance.

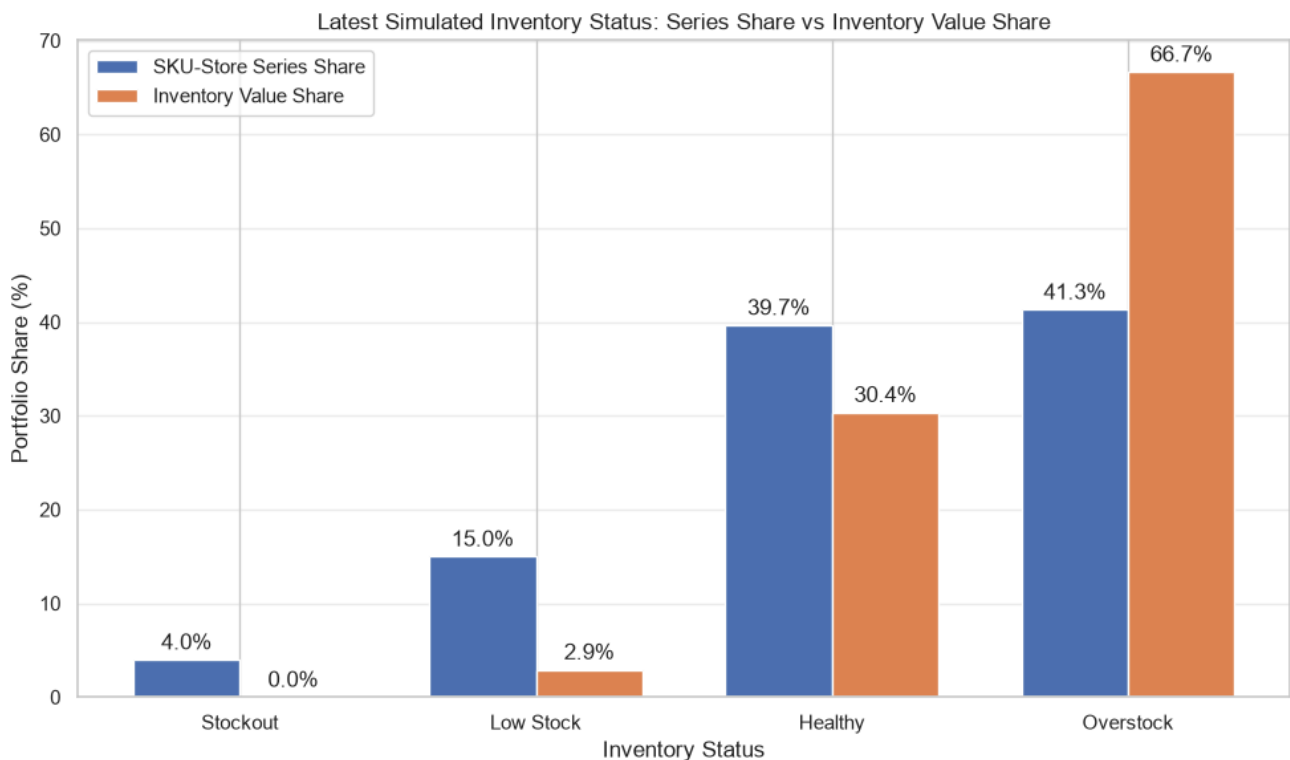
	Inventory Status	SKU-Store Series	Series Share (%)	Simulated Closing Stock	Simulated On-Order Units	Estimated Inventory Value	Inventory Value Share (%)	Median Coverage Days	Median Recent Daily Demand	Median Days Since Last Sale
0	Stockout	12	4.00	0	226	0.00	0.00	0.00	0.52	0.00
1	Low Stock	45	15.00	194	1036	416.36	2.94	5.87	0.53	0.00
2	Healthy	119	39.67	1649	0	4,304.61	30.38	15.47	0.66	1.00
3	Overstock	124	41.33	4641	0	9,449.58	66.68	26.62	0.64	1.00

✅ Latest inventory status and coverage prepared.

SKU-store series analysed : 300

Inventory statuses : 4

Coverage range : 0.00 to 565.20 days



✅ Inventory-status figure created.

Figure saved to: C:\Users\hp\Project FORESIGHT\reports\figures\11_2_inventory_status_and_value_share.png

Simulated Inventory Status and Coverage Findings

- **Stockout exposure: 12 SKU-store series (4.00%)** had zero simulated closing stock. These series had **226 units on order**, indicating simulated replenishment activity.
- **Low-stock exposure: 45 series (15.00%)** were classified as Low Stock, with median inventory coverage of **5.87 days**.
- **Healthy inventory: 119 series (39.67%)** were classified as Healthy and represented **30.38% of estimated simulated inventory value**.
- **Overstock concentration: 124 series (41.33%)** were classified as Overstock but accounted for **66.68% of estimated simulated inventory value**.

- **Coverage pattern:** Median coverage increased from **0 days for Stockout** to **5.87 days for Low Stock**, **15.47 days for Healthy**, and **26.62 days for Overstock**.
- **Coverage caution:** The maximum coverage of **565.20 days** is likely influenced by extremely low recent demand and should be interpreted together with demand intermittency in Section 11.3.
- **Planning implication:** The simulated portfolio shows inventory imbalance at both ends—limited availability for some series and substantial value concentration in overstocked series.

These results describe the latest simulated inventory snapshot and are not final stockout-risk, overstock-risk, or replenishment recommendations.

✓ Section 11.2: Simulated Inventory Status and Coverage is complete.

11.3 Demand Variability and Intermittency

Historical SKU-store demand behaviour will be evaluated using:

- average daily demand;
- demand standard deviation;
- coefficient of variation;
- active selling-day frequency;
- zero-demand frequency;
- recent demand and sales recency.

These measures will be used to distinguish relatively stable demand from variable, volatile, and intermittent patterns.

Thresholds will be selected from the observed distributions and documented before classification.

Demand variability should be interpreted together with selling frequency because a high coefficient of variation may reflect intermittent sales rather than continuous volatility.

✓ New demand-variability measures calculated.

SKU-store series analysed : 300

Historical date range : 2011-01-29 to 2016-04-24

New measures created : average demand, standard deviation, coefficient of variation, demand-change frequency

	Demand Behaviour Measure	Series Count	Mean	Standard Deviation	Minimum	25th Percentile	Median	75th Percentile	90th Percentile	95th Percentile	Maximum
0	average_daily_demand	300.00	1.00	1.54	0.01	0.26	0.48	1.05	1.93	4.73	9.78
1	demand_standard_deviation	300.00	1.47	1.76	0.14	0.63	0.95	1.52	2.43	5.02	11.21
2	coefficient_of_variation	300.00	2.14	1.01	0.69	1.46	1.99	2.57	3.23	3.86	10.60
3	demand_change_frequency_pct	300.00	41.43	18.72	1.26	25.85	41.40	58.51	66.11	71.35	86.66
4	full_history_selling_frequency_pct	300.00	33.63	19.21	1.05	17.59	30.48	47.94	63.64	66.36	86.30
5	recent_selling_frequency_pct	300.00	39.58	22.62	1.11	21.11	37.78	55.83	70.11	80.00	96.67
6	recent_zero_demand_frequency_pct	300.00	60.42	22.62	3.33	44.17	62.22	78.89	90.00	93.33	98.89
7	recent_units_sold	300.00	82.91	104.04	2.00	26.75	56.50	101.25	170.10	234.20	953.00
8	days_since_last_sale	300.00	6.85	18.97	0.00	0.00	1.00	3.00	10.10	76.10	86.00

✅ Demand-behaviour distributions prepared.

SKU-store series reviewed : 300

Measures reviewed : 9

Next step : select documented segmentation thresholds

	Segmentation Rule	Threshold Basis	Threshold Value
0	Very Low / Inactive Demand	Days since last sale $\geq$ 95th percentile	76.10
1	Intermittent Demand	Recent selling frequency $\leq$ 25th percentile	21.11
2	Stable Demand	Coefficient of variation $\leq$ 25th percentile	1.46
3	Volatile Demand	Coefficient of variation $>$ 75th percentile	2.57
4	Variable Demand	Series not captured by the other rules	NaN

	Demand Behaviour Segment	SKU-Store Series	Series Share (%)
0	Stable Demand	65	21.67
1	Variable Demand	122	40.67
2	Volatile Demand	34	11.33
3	Intermittent Demand	64	21.33
4	Very Low / Inactive Demand	15	5.00

✅ Demand-behaviour segments assigned.

Classified series : 300

Segments created : 5

	Demand Behaviour Segment	SKU-Store Series	Series Share (%)	Median Average Daily Demand	Median Coefficient of Variation	Median Recent Selling Frequency (%)	Median Recent Zero-Demand Frequency (%)	Median Recent Units Sold	Median Days Since Last Sale
0	Stable Demand	65	21.67	1.39	1.25	63.33	36.67	130.00	0.00
1	Variable Demand	122	40.67	0.52	1.91	43.33	56.67	67.50	0.00
2	Volatile Demand	34	11.33	0.18	2.97	29.44	70.56	35.50	1.00
3	Intermittent Demand	64	21.33	0.21	2.67	13.33	86.67	16.50	3.00
4	Very Low / Inactive Demand	15	5.00	0.66	2.16	5.56	94.44	12.00	81.00

✓ Demand-behaviour segment profiles prepared.

Segments summarised : 5

Series represented : 300

Demand Variability and Intermittency Findings

- **Stable demand: 65 SKU-store series (21.67%)** showed the lowest relative variability, with a median coefficient of variation of **1.25**. These series also recorded the highest median recent selling frequency at **63.33%**.
- **Variable demand: 122 series (40.67%)** formed the largest segment. Their median coefficient of variation was **1.91**, with demand recorded on **43.33%** of recent days.
- **Volatile demand: 34 series (11.33%)** showed a median coefficient of variation of **2.97**, indicating large fluctuations relative to their average demand.
- **Intermittent demand: 64 series (21.33%)** recorded sales on only **13.33%** of recent days and had a median zero-demand frequency of **86.67%**.
- **Very low or recently inactive demand: 15 series (5.00%)** had a median of **81 days since the latest sale** and a recent selling frequency of only **5.56%**.
- **Forecasting implication:** Stable series may support conventional forecasting methods, while volatile and intermittent series will require greater uncertainty allowances or specialised forecasting treatment.
- **Interpretation caution:** The behaviour segments are relative portfolio classifications based on observed distribution thresholds. They are separate from the product movement classes developed in Section 9.

✓ Section 11.3: Demand Variability and Intermittency is complete.

11.4 Inventory–Demand Planning Segmentation

Inventory conditions and demand behaviour will be combined into practical planning segments.

Examples may include:

- fast or stable demand with limited inventory;
- volatile demand requiring greater uncertainty allowance;
- intermittent demand requiring specialised forecasting treatment;
- weak movement with comparatively high inventory;
- adequate inventory with no immediate concern.

The weak-movement watchlist created in Section 9 will be reused rather than recalculated.

These segments are preliminary analytical labels. They are not final stockout-risk, overstock-risk, or replenishment recommendations.

	Inventory-Demand Planning Segment	SKU-Store Series	Series Share (%)
0	Overstock / Active-Demand Review	84	28.00
1	Balanced / Routine Monitoring	80	26.67
2	Availability Priority	47	15.67
3	Overstock / Weak-Demand Review	40	13.33
4	Healthy Stock / Forecasting Attention	39	13.00
5	Limited Stock / Irregular Demand	10	3.33

✓ Inventory-demand planning segments assigned.

Classified series : 300

Segments created : 6

Section 9 watchlist flags reused: 63

	Inventory-Demand Planning Segment	SKU-Store Series	Series Share (%)	Estimated Inventory Value	Inventory Value Share (%)	Median Coverage Days	Median Recent Daily Demand	Median Recent Selling Frequency (%)	Median Days Since Last Sale	Section 9 Watchlist Series
0	Availability Priority	47	15.67	381.75	2.69	3.38	0.74	43.33	0.00	0
1	Limited Stock / Irregular Demand	10	3.33	34.61	0.24	8.53	0.18	13.89	0.00	5
2	Balanced / Routine Monitoring	80	26.67	3,424.27	24.16	12.71	0.90	50.00	0.00	0
3	Healthy Stock / Forecasting Attention	39	13.00	880.34	6.21	30.00	0.19	13.33	4.00	26
4	Overstock / Active-Demand Review	84	28.00	6,357.90	44.87	24.28	0.84	48.89	0.50	0
5	Overstock / Weak-Demand Review	40	13.33	3,091.68	21.82	59.01	0.20	12.22	8.00	32

✓ Inventory-demand planning profiles prepared.

Planning segments summarised : 6

Series represented : 300

Section 9 watchlist reused : 63

Inventory–Demand Planning Findings

- **Availability concern: 47 series** had active demand but only **3.38 median coverage days**.
- **Balanced position: 80 series** showed comparatively stable inventory and demand conditions.
- **Overstock concern: 40 weak-demand series** held **21.82% of simulated inventory value** with **59.01 median coverage days**.
- **Active overstock: 84 series** held **44.87% of inventory value**, but continued demand means they should be reviewed rather than automatically marked down.
- All **63 Section 9 watchlist series** were reused without recalculation.

These are descriptive EDA segments. Final risk scores and recommendations will be developed in Notebook 07.

✓ Section 11.4: Inventory–Demand Planning Segmentation is complete.

11.5 Forecasting and Inventory Implications

This subsection consolidates the most useful inventory and demand-behaviour findings.

The final summary will document:

- inventory-status distribution;
- coverage concerns;
- stable, variable, volatile, and intermittent demand counts;
- important combined inventory-demand patterns;
- forecasting treatment;
- inventory-planning implication;
- limitation or caution.

Only one concise findings table and one short Markdown interpretation will be included.

✓ Forecasting and inventory implications prepared.

Planning segments covered : 6

✓ Section 11.5 completed: Forecasting and inventory implications consolidated.

Forecasting and Inventory Implications

- Stable and balanced series can proceed through standard forecasting evaluation.
- Volatile and intermittent series require uncertainty-aware or specialised forecasting treatment.
- Limited inventory should be assessed against future lead-time demand before replenishment decisions.
- Overstock cases must be separated between active-demand and weak-demand products.
- Final risk scores, reorder quantities, markdown actions, and financial exposure belong to Notebook 07.

Inventory findings remain simulation-based and descriptive.

✓ Section 11.5: Forecasting and Inventory Implications is complete.

✓ Section 11: Inventory and Demand Behaviour is complete.

12. Business Insights, Recommendations, and EDA Insight Memo

This section converts the strongest validated EDA findings into prioritised business recommendations and prepares the stakeholder-facing EDA insight memo.

Only distinct, evidence-based findings will be retained. Recommendations remain exploratory and will not replace the forecast-based risk scoring and inventory decisions developed in Notebook 07.

Objectives

- consolidate the strongest findings from Sections 7–11;
- connect each finding to a practical business response;
- prioritise actions by business relevance and feasibility;
- disclose simulation, estimation, and analytical limitations;
- prepare the required EDA insight memo for non-technical stakeholders.

Analysis Structure

- **12.1:** Build one evidence-based insight and recommendation register.
- **12.2:** Consolidate assumptions and scope boundaries.
- **12.3:** prepare and export the EDA insight memo.

Final reorder quantities, risk scores, markdown decisions, and financial exposure will be developed in Notebook 07.

12.1 Final Insight and Recommendation Register

The strongest findings from demand, seasonality, business performance, product movement, demand drivers, and simulated inventory analysis will be consolidated into one decision-oriented register.

Each entry will contain:

- **Business Insight**
- **Supporting Evidence**
- **Business Significance**
- **Recommended Response**
- **Priority**
- **Analytical Source**
- **Limitation or Scope Note**

Approximately six to eight distinct insights will be retained. Earlier analytical tables will be reused rather than recalculated.

	Insight Area	Priority	Analytical Source
0	Demand Timing	Near Term	Section 7
1	Category Performance	Strategic	Section 8
2	Product Concentration	Immediate	Sections 8–9
3	Store and Geography	Near Term	Section 8
4	Demand Drivers	Monitor	Section 10
5	Demand Behaviour	Near Term	Sections 9 and 11
6	Simulated Inventory	Immediate	Section 11

✓ Final insight and recommendation register prepared.

Distinct insights retained : 7

Priority levels represented : 4

Final Insight Register Summary

- **Immediate priorities:** protect high-contribution products and review simulated weak-demand overstock.
- **Near-term priorities:** incorporate weekly seasonality, geographic differences, and intermittent-demand treatment into later modelling.
- **Monitoring priority:** validate price, promotion, event, and SNAP features through time-based forecasting tests.
- **Strategic priority:** evaluate category performance using both demand volume and revenue.

These recommendations are based on historical EDA. Final forecast-based inventory actions will be developed in Notebook 07.

✓ Section 12.1: Final Insight and Recommendation Register is complete.

12.2 EDA Insight Memo

A concise stakeholder-facing memo will summarise the strongest findings and recommendations from Notebook 02.

The memo will include:

- business objective and analytical scope;
- key data-quality and integration notes;
- demand, seasonality, and product-movement findings;
- demand-driver and simulated inventory observations;
- prioritised recommendations;
- assumptions and limitations.

The planned output is:

reports/executive_report/EDA_Insight_Memo.pdf

The memo will clearly identify the 300-series development scope, simulated inventory, estimated financial measures, derived promotion indicators, and association-based findings.

	Memo Component	Items Prepared
0	Business and Scope Metadata	8
1	Insights and Recommendations	7
2	Assumptions and Limitations	6

✅ EDA insight memo content prepared.

Metadata items : 8

Insights included : 7

Limitations included : 6

Next step : generate and validate the stakeholder PDF

✅ EDA insight memo PDF generated successfully.

PDF path : C:\Users\hp\Project FORESIGHT\reports\executive_report\EDA_Insight_Memo.pdf

File size: 6.40 KB

	Required Section	Occurrences
0	Executive Overview	1
1	Key Findings and Recommendations	1
2	Assumptions and Limitations	1
3	Next Analytical Steps	1

✅ Final EDA insight memo validation passed.

Page count : 3

Insight areas found : 7 of 7

Duplicate sections : 0

EDA Insight Memo Summary

- The stakeholder memo consolidates the **7 strongest EDA insights and recommendations** from Notebook 02.
- The final PDF contains all required sections exactly once and passed structural validation.
- The memo is **3 pages**, concise, business-facing, and clearly distinguishes simulated, estimated, and association-based findings.
- Final forecast-driven inventory actions remain reserved for Notebook 07.

Output: reports/executive_report/EDA_Insight_Memo.pdf

✅ Section 12.2: EDA Insight Memo is complete.

✅ Section 12: Business Insights, Recommendations, and EDA Insight Memo is complete.

13. Export and Final Validation

This section consolidates the final outputs produced throughout Notebook 02, exports only the remaining required deliverables, and performs a final validation audit.

Previously created datasets, figures, summaries, and reporting outputs will be reused and verified rather than recalculated or regenerated.

Objectives

- register the validated Notebook 02 outputs;
- export only the remaining required analytical tables;
- verify previously generated datasets, figures, and reports;
- confirm file existence, naming, location, and non-empty content;
- perform the final Notebook 02 analytical and deliverable audit.

Export Structure

- **13.1:** Register final Notebook 02 outputs.
- **13.2:** Export remaining required outputs.
- **13.3:** Validate all outputs and complete the final notebook audit.

13.1 Register Final Notebook Outputs

This subsection defines the final Notebook 02 deliverables using validated objects and files created earlier in the analysis.

No analytical results are recalculated or redisplayed here. Existing outputs are reused directly to preserve consistency and avoid repetition.

	Output	Object / File		Location	Status
0	Integrated analytical dataset	analytics_base_daily	data/interim/analytics_base_daily.parquet		Created earlier
1	Executive KPI summary	eda_kpi_summary	reports/tables/eda_kpi_summary.csv		Created earlier
2	Movement classification summary	movement_class_summary	reports/tables/movement_class_summary.csv		Pending export
3	Demand-driver screening	demand_driver_feature_screening	reports/tables/demand_driver_feature_screening...		Pending export
4	Inventory planning summary	inventory_demand_planning_summary	reports/tables/inventory_planning_summary.csv		Pending export
5	Business insight summary	final_insight_recommendation_register	reports/tables/business_insight_summary.csv		Pending export
6	EDA Insight Memo	EDA_Insight_Memo.pdf	reports/executive_report/EDA_Insight_Memo.pdf		Created earlier
7	Analytical figures	Validated figures	reports/figures/		Created earlier

✅ Final Notebook 02 output register prepared.
Registered deliverables: 8

13.2 Export Remaining Required Outputs

Only the analytical tables that have not already been saved earlier in Notebook 02 will be exported here.

Previously generated datasets, figures, KPI summaries, and the EDA Insight Memo will not be regenerated.

	File	Rows	Columns	Size (KB)
0	movement_class_summary.csv	3	15	0.90
1	demand_driver_feature_screening.csv	4	6	1.10
2	inventory_planning_summary.csv	6	12	1.12
3	business_insight_summary.csv	7	7	2.32

✔ Remaining Notebook 02 tables exported successfully.
Files exported: 4
Output directory: C:\Users\hp\Project FORESIGHT\reports\tables

13.3 Final Output Validation and Notebook Audit

A consolidated audit verifies that the required Notebook 02 deliverables were created successfully and that the major analytical components remain available for downstream use.

The final validation checks:

- existence and non-empty content of required exported files;
- availability of validated analytical figures;
- successful generation of the EDA Insight Memo;
- completion of trend and seasonality analysis;
- completion of business-performance and movement analysis;
- completion of demand-driver and inventory-demand analysis;
- availability of the final insight and recommendation register.

The notebook will proceed to its conclusion only after all critical audit checks pass.

The complete Notebook 02 PDF will be exported after the Notebook Conclusion is finished and the notebook is saved in its final state.

	Output	Exists	File Size (KB)	Figure Count	Status
0	Integrated analytical dataset	True	4,708.92	<NA>	PASS
1	Executive KPI summary	True	2.46	<NA>	PASS
2	Movement classification summary	True	0.90	<NA>	PASS
3	Demand-driver screening	True	1.10	<NA>	PASS
4	Inventory planning summary	True	1.12	<NA>	PASS
5	Business insight summary	True	2.32	<NA>	PASS
6	EDA Insight Memo	True	6.40	<NA>	PASS
7	Analytical figures	True	<NA>	20	PASS

	Audit Item	Status
0	Trend and seasonality analysis	PASS
1	Business performance analysis	PASS
2	Movement classification	PASS
3	Demand-driver analysis	PASS
4	Inventory-demand analysis	PASS
5	Final insights and recommendations	PASS
6	EDA Insight Memo	PASS

✅ Final Notebook 02 audit passed.

Required outputs validated : 8

Analytical areas validated : 7

Figures available : 20

Final Audit Summary

- **8 required output groups** passed file validation.
- **7 major analytical areas** passed the completion audit.
- **20 validated analytical figures** are available in `reports/figures/`.
- The **EDA Insight Memo** was successfully generated and validated.
- Notebook 02 analytical outputs are ready for finalisation and downstream feature engineering.

✅ Section 13: Export and Final Validation is complete.

✅ 14. Notebook Conclusion

Notebook 02 completed the exploratory analysis of Project FORESIGHT and transformed the validated datasets from Notebook 01 into evidence-based business insights for demand planning, forecasting preparation, and simulated inventory review.

The notebook analysed demand patterns, business performance, product movement, demand drivers, and inventory behaviour; produced stakeholder-facing outputs; and passed the final deliverable audit.

Conclusion Structure

- **14.1:** Summarise the work completed and key analytical outcomes.
- **14.2:** Confirm final outputs and important scope boundaries.
- **14.3:** Confirm readiness for Notebook 03 — Feature Engineering.

14.1 Work Completed and Key Outcomes

Notebook 02 completed the full exploratory analysis workflow for the validated Project FORESIGHT datasets.

Work Completed

- built and validated the integrated daily analytical base;
- calculated executive demand, revenue, and simulated inventory KPIs;
- analysed trend, seasonality, weekday, weekend, monthly, quarterly, and yearly demand behaviour;
- evaluated category, store, state, product, and SKU-store performance;
- identified demand concentration, movement classes, weak-demand series, and dead-stock conditions;
- screened price, promotion, event, and SNAP relationships;
- analysed demand variability, intermittency, inventory coverage, and planning segments;
- consolidated the strongest findings into a prioritised business recommendation register;
- generated and validated the stakeholder-facing EDA Insight Memo.

Key Outcomes

- demand shows meaningful temporal and portfolio concentration patterns;
- product and geographic performance vary materially across the development scope;
- demand-driver relationships are mixed and require time-based forecasting validation;
- intermittent and weak-demand series require different forecasting treatment from stable series;
- simulated inventory exposure is concentrated in overstock and weak-demand positions.

Notebook 02 therefore provides the analytical foundation required for leakage-safe feature engineering and forecasting in the next stage.

14.2 Final Outputs and Scope Boundaries

Final Outputs

Notebook 02 produced and validated the following major deliverables:

- integrated daily analytical dataset;
- executive KPI summary;
- movement classification summary;
- demand-driver screening summary;
- inventory planning summary;
- business insight and recommendation summary;
- 20 validated analytical figures;
- 3-page stakeholder-facing EDA Insight Memo.

Scope Boundaries

The findings should be interpreted within the following project constraints:

- the M5 dataset is adapted to the fictional NorthBay Living business case;

- the analysis uses the approved 300 SKU-store development scope;
- inventory records are simulated;
- cost, margin, and inventory-value measures are estimated or assumption-based;
- promotion indicators are derived and have limited coverage;
- demand-driver relationships are associative rather than causal;
- final forecast-based stockout risk, overstock risk, reorder quantities, markdown actions, and financial exposure remain outside Notebook 02.

These boundaries preserve a clear distinction between exploratory evidence and the forecasting and inventory-decision logic developed in later notebooks.

14.3 Readiness for Notebook 03

Notebook 02 has satisfied its exploratory-analysis objectives and passed the final output audit.

Readiness Confirmation

- validated processed datasets were successfully integrated for EDA;
- required demand, seasonality, business-performance, movement, driver, and inventory analyses were completed;
- business insights and recommendations are supported by validated evidence;
- required analytical tables, figures, and the EDA Insight Memo were generated successfully;
- assumptions, simulation boundaries, and analytical limitations were documented;
- all critical Notebook 02 output checks passed.

Next Step

Project FORESIGHT will now proceed to:

`03_Feature_Engineering.ipynb`

Notebook 03 will transform the validated analytical data into **leakage-safe forecasting features**, including historical demand, calendar, pricing, promotion, lag, rolling, product, store, and demand-behaviour features.

Final Status: Notebook 02 — Exploratory Data Analysis completed successfully.

- ✓ Section 14: Notebook Conclusion is complete.
- ✓ Notebook 02: Exploratory Data Analysis is complete.